Cooperative Adaptive Cruise Control for Connected Autonomous Vehicles by Factoring Communication-Related Constraints☆

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Abstract

Emergent cooperative adaptive cruise control (CACC) strategies being proposed for platoon formation in the connected autonomous vehicle (CAV) context mostly assume idealized fixed information flow topologies (IFTs) for the platoon, implying guaranteed vehicle-to-vehicle (V2V) communications for the IFT assumed. In reality, V2V communications are unreliable due to failures resulting from communication-related constraints such as interference and information congestion. Since CACC strategies entail continuous information broadcasting, communication failures can occur in congested CAV traffic networks, leading to a platoon's IFT varying dynamically. To explicitly factor IFT dynamics and to leverage it to enhance the performance of CACC strategies, this study proposes the idea of dynamically optimizing the IFT for CACC, labeled the CACC-OIFT strategy. Under CACC-OIFT, the vehicles in the platoon cooperatively determine in real-time which vehicles will dynamically deactivate or activate the "send" functionality of their V2V communication devices to generate IFTs that optimize the platoon performance in terms of string stability under the ambient traffic conditions. The CACC-OIFT consists of an IFT optimization model and an adaptive Proportional-Derivative (PD) controller. Given the adaptive PD controller with a two-predecessor-following scheme, and the ambient traffic conditions and the platoon size just before the start of a time period, the IFT optimization model determines the optimal IFT that maximizes the expected string stability in terms of the energy of speed oscillations. This expectation is because each IFT has specific degeneration scenarios whose probabilities are determined by the communication failure probabilities for that time period based on the ambient traffic conditions. The optimal IFT is deployed for that time period, and the adaptive PD controller continuously determines the car-following behaviors of the vehicles based on the unfolding degeneration scenario for each time instant within that period. The effectiveness of the proposed CACC-OIFT is validated through numerical experiments in NS-3 based on NGSIM field data. The results indicate that the proposed CACC-OIFT can significantly enhance the string stability of platoon control in an unreliable V2V communication context, outperforming CACCs with fixed IFTs or with passive adaptive schemes for IFT dynamics.

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Keywords: CACC; Communication failure; Optimization; String stability; Dynamic information flow topology.

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1. Introduction

Traffic oscillation refers to "stop and go" traffic propagation (Li et al., 2010), which can cause uncomfortable driving experience, safety problems, additional energy consumption, and negative environmental impacts. A widely-adopted method to deal with traffic oscillations is adaptive cruise control (ACC), which controls the speed in pace with the preceding vehicles. The advent of vehicle-to-vehicle (V2V) communications using dedicated short-range communication (DSRC) technologies is enabling vehicles to receive additional information from other connected vehicles and the infrastructure. This will provide more opportunities for connected autonomous vehicles (CAVs) to enhance their situational awareness and performance through the implementation of more robust system-level vehicle control strategies, especially platoon-based cooperative adaptive cruise control (CACC) (Nieuwenhuijze et al., 2012).

CACC is an extension of ACC, used to minimize speed differences among vehicles in a platoon and maintain stable and safe headways between adjacent vehicles (Zhou et al., 2017). The CACC literature, discussed hereafter, assumes a platoon with pure (100%) CAVs. Typically, a CACC framework has four components (Li et al., 2015): (i) *node dynamics* (ND), which describes the dynamics of each vehicle in the platoon, such as second-order models (Wang et al., 2014), or third-order models (Guo et al., 2012); (ii) *vehicle-level information flow topology* (VIFT) for a CAV, which describes the configuration of V2V communication links from one CAV to one or more CAVs in the platoon, for example, predecessor-following-leader (Naus et al., 2010) and two-predecessor-following (Zheng et al., 2017) VIFT schemes; (iii) *decentralized controller*, which uses information from other vehicles in the platoon to implement control strategies, such as Proportional-Integral-Derivative (PID) controller (Swaroop et al., 1996), sliding mode controller (Gao et al., 2018), and model predictive controller (Wang et al., 2014; Zhou et al., 2017); and (iv) *formation geometry*, which describes the desired headway between vehicles. Recent studies have modeled the four components of a CAV platoon in different ways, such as the constant distance (CD) policy (Gong et al., 2016), and the constant time headway (CTH) policy (Zhou et al., 2017).

Among the four CACC framework components, the VIFT is closely related to the status of V2V communications. Almost all existing CACC studies assume identical VIFTs for all vehicles in a platoon. As discussed hereafter, consistent with the real world, our study does not constrain the VIFTs to be identical. We label the information flow topology at the platoon level as "IFT," which illustrates the configuration of V2V communication links of all vehicles in the platoon at any time instant. By introducing the time dimension, we consider the inherent dynamics of IFT, further enhancing modeling realism. By contrast, most studies using a CACC design assume an idealized predetermined, fixed IFT. This assumption ignores the fact that the IFT (and by implication, the VIFTs) can change dynamically due to V2V communication failures (Gao et al., 2018; Talebpour et al., 2016). A communication failure may occur due to communication interference or information congestion (Kim et al., 2017; Wang et al., 2018), especially when the ambient (pure CAV) traffic is congested. Information congestion is the reduced quality of service when a communication network node carries more data than it can handle, which in the context of V2V communications is modeled through the potential for failure of information propagation in a V2V communications-enabled traffic network (Wang et al., 2018). Communication interference typically refers to the disruption of a signal as it travels between a sender and a receiver. In the CACC platoon context, though the transmission distance is close enough, interference can arise due to the ambient traffic conditions. A critical reason for interference from other vehicles is the mechanism of the DSRC protocol defined by IEEE 802.11p. In telecommunications, information is transmitted via channels. If more than one sender tries to send information via the same channel at the same time, it will cause interference for both vehicles, and the resulting information collision can cause the transmission to fail. To reduce the probability of information collision, IEEE 802.11p inherits the contention mechanism from IEEE 802.11, which requires every sender to compete for the sending chance. Thereby, the probability of information collision is decreased through this mechanism. However, for V2V communications, if a sender fails to win a sending chance before a new message is generated, the old message will be dropped, and is counted as a communication failure (Qiu et al., 2015). Other factors like the hidden node effect and capture effect can also cause communication failures. In all such cases, the distance between some senders is too large for them to sense each other or to keep transmitting signals with enough magnitude so that they can be successfully received. Hence, sender-based communication failure is addressed in this study. Receiver-related failure is not considered because the spacing between vehicles in a platoon is small enough that a message can be received as long as it is sent.

If communication failures occur, a CACC with a fixed IFT (CACC-FIFT) may execute an erroneous control action or degrade to adaptive cruise control (ACC), which diminishes platoon performance related to mobility, stability, and even safety. To mitigate the negative effects of communication failure, a few studies have proposed novel CACC strategies for a pure CAV platoon by considering dynamic IFT degeneration scenarios. Here, a degeneration scenario

for a given IFT refers to any configuration with one or more link communication failures for that IFT. Gong et al. (2019) propose a CACC strategy with dynamic IFT degeneration scenarios (CACC-DIFT), in which a PD controller combined with an acceleration feedforward filter is designed to counteract the IFT dynamics in the platoon. Depending on the IFT degeneration scenario that unfolds at different time instants, the CACC will change the controller parameters to maintain string stability performance rather than degrade to ACC. Gao et al. (2018) design a distributed sliding mode controller based on a linear matrix inequality method to ensure string stability under uncertain but eigenvalue-bounded IFT degeneration scenarios. However, both these studies consider IFT dynamics passively, implying that the controller uses only the functioning links when others have V2V communication failures. While such passive approaches may improve the control performance under unreliable V2V networks, their performance is constrained by ambient traffic conditions which determine the communication failure probabilities of the various IFT links. That is, when the ambient traffic density is higher, these probabilities are higher which indicates the potential lower robustness of the IFT. Further, studies (Hafeez et al. 2013) suggest that the IFT in terms of the number of vehicles in the platoon with activated "send" functionality of V2V devices within the communication range, is another key factor that impacts communication failures. That is, these failure probabilities are higher if several CAVs within communication range have their "send" functionality activated, which inspires the key idea of this study that communication failure probabilities can be reduced (i.e., the robustness of the IFT can be improved in terms of the communication reliability) by deactivating the "send" functionality of a subset of CAVs. However, we cannot deactivate the "send" functionality of all CAVs in a platoon as it will diminish the platoon's awareness capability. If all CAVs deactivate their "send" functionality, the IFT is at its most robust in that no communication failures will occur. Then, all CAVs will operate in the ACC mode which leads to poor theoretical performance, implying that the expected performance will not be good. By contrast, the best string stability under perfect communication conditions is achieved when all CAVs activate their "send" functionality, though the probability of communication failure is also at its highest (i.e. low robustness of IFT), indicating poor robustness for platoon control. Then, the expected string stability performance is also not good. Hence, there is a tradeoff between the theoretical string stability of an IFT and its robustness. To account for these real-world characteristics, our study proposes the novel idea of proactively controlling in real time the number of platoon vehicles with "send" functionality activated based on the unfolding ambient traffic conditions so as to enhance communication reliability, with the objective of maximizing platoon performance in terms of string stability. We label this strategy the CACC with dynamically optimized IFT, or CACC-OIFT, which aims to achieve this tradeoff and enhance the expected platoon performance in terms of the string stability with communication constraints.

Enabling the CACC-OIFT strategy entails addressing some key challenges. First, to account for the time-varying nature of communication failures, all degeneration scenarios of an IFT and their probabilities should be determined. The probability of each degeneration scenario depends on the probability of the communication failure of each link in the IFT, which is itself dynamic and depends on the unfolding traffic conditions. Second, the platoon control performance in terms of string stability needs to be theoretically formulated in the expected sense over all degeneration scenarios for an IFT. While the existing literature uses simulation-based methods (Swaroop et al., 1996; Schakel et al., 2010; Nieuwenhuijze et al., 2012; Zhou et al., 2017) to numerically determine control performance, it is difficult to integrate such approaches in a rigorous optimization model. Third, an adaptive controller is needed to control the car-following behaviors of the vehicles in the CAV platoon based on the unfolding degeneration scenarios for the optimal IFT at different time instants. Hence, there is the need to factor IFT dynamics while ensuring string stability.

The proposed CACC-OIFT strategy for a time period seeks to determine the optimal IFT that maximizes the expected string stability performance to damp traffic oscillations by deactivating or activating the "send" functionality of the V2V communication devices of the vehicles in the platoon, and deploys it for every time instant within that period based on the unfolding degeneration scenarios for that IFT due to V2V communication failures. It includes an IFT optimization model and an adaptive Proportional-Derivative (PD) controller. Fig. 1 illustrates the conceptual flowchart of CACC-OIFT and its operational deployment. Fig. 1(a) shows the various components of CACC-OIFT and their linkages in the time dimension. Given the ambient traffic conditions and platoon size at some time instant $\Delta\tau$ before the start of time period τ (the period from time instant t_τ to $t_{\tau+1}$ (i.e., $[t_\tau, t_{\tau+1}]$) in Fig. 1(b)), the IFT optimization model first identifies the candidate IFTs corresponding to the platoon size and their degeneration scenarios. Note that the set of all possible IFTs and their degenerations scenarios is determined offline as they are time-invariant and can be predetermined. The subset of IFTs corresponding to the current platoon size denotes the candidate IFT set. Second, the ambient traffic conditions are used to determine the probabilities of the degeneration scenarios for each candidate IFT as these traffic conditions determine the V2V (link) communication failure

probabilities. Third, the string stability for each degeneration scenario for each candidate IFT is obtained from the predetermined string stabilities for all degeneration scenarios for all possible IFTs, computed offline using the transfer function in frequency domain of the given adaptive PD controller. This study uses the speed oscillation energy of the platoon as an indicator of string stability performance, which treats the speed oscillation as a signal and computes the sum of speed oscillation energies of all vehicles in the platoon in frequency domain. A lower value for this sum implies that traffic oscillations are damped as they propagate through the platoon, implying better string stability. The optimal IFT (activations and deactivations of "send" functionalities for the platoon vehicles) for time period τ is determined as the candidate IFT which has the maximum expected string stability across all of its degeneration scenarios.

As shown in Fig. 1(b), the IFT optimization model will determine the optimal IFT for time period τ at time instant $t_\tau - \Delta\tau$ as it takes $\Delta\tau$ time to solve for the optimal IFT. Note that it is assumed that the ambient traffic conditions and platoon size do not vary within the time period. From Fig. 1(a), the operational deployment of CACC-OIFT starts by proactively deploying the optimal IFT for the first time instant t_τ of period τ. However, due to V2V communication failures, different time instants in period τ can have a different degeneration scenario of the optimal IFT manifest. The adaptive PD controller continuously determines the car-following behaviors of the vehicles based on the unfolding degeneration scenario for each time instant (i.e., t_τ, $t_\tau + 1$,…, $t_{\tau+1}$) in period τ, thereby controlling vehicular location and dynamics. As shown in Fig. 1(b), at $t_{\tau+1} - \Delta\tau$, the IFT optimization model will update the optimal IFT for the next time period $\tau + 1$. This process continues for the time horizon of interest.

In CACC-OIFT implementation, after the leading vehicle receives information on ambient traffic conditions at $t_\tau - \Delta\tau$, it distributes computational tasks to the other platoon vehicles which perform these computations in parallel to enable computational efficiency. Then, the leading vehicle collects the results of the computations and determines the optimal IFT to deploy in the next cycle. This ensures that the computing process is completed within $\Delta\tau$. Note that the length of each time period τ is flexible; however, we only update the optimal IFT when the platoon size or the ambient traffic state related to the traffic oscillations changes. According to Li et al. (2010), the time-invariant pattern of traffic oscillations usually lasts more than 10 minutes, suggesting possible values for the update frequency.

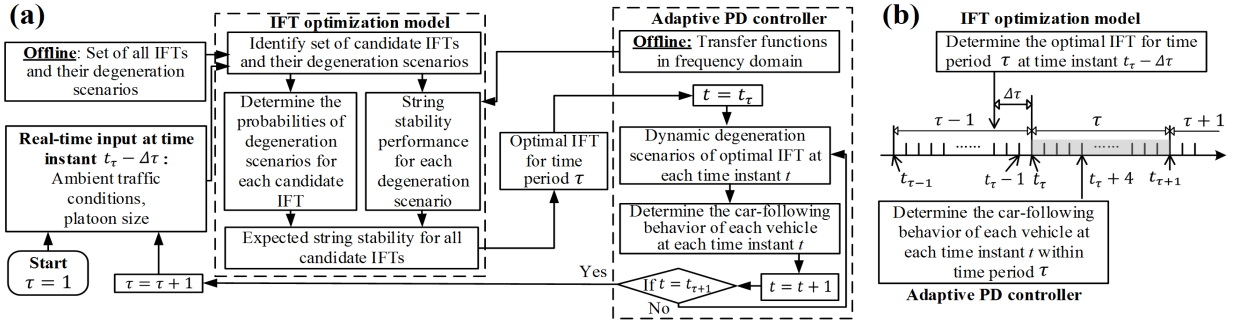


Fig. 1. (a) Conceptual flowchart of CACC-OIFT; (b) Operational deployment of CACC-OIFT.

As the IFT optimization model optimizes the IFT for each time period τ and the adaptive PD controller controls the car-following behaviors for each time instant t, we will formulate the IFT optimization model for one time period in Section 3 and the adaptive PD controller for a time instant within that time period in Section 4. Hence, for notational convenience, we will omit τ and t in these sections. A two-step algorithm is developed to solve the IFT optimization problem, and several critical properties are proved; for example, the leading vehicle in the platoon should always activate its "send" functionality. The effectiveness of the proposed CACC is validated using NGSIM field data (US DOT, 2007) in network simulator NS-3. The results reveal that the algorithm can solve the IFT optimization model for a platoon of considerable size (15 CAVs) in a practically deployable time duration (less than a minute). The proposed CACC-OIFT can significantly damp traffic oscillations and enhance string stability in an unreliable V2V communications context, outperforming CACCs with fixed IFTs or with passive adaptive schemes for IFT dynamics.

The major contributions of the paper are as follows:

(1). We propose an IFT optimization model to explicitly factor the inherent IFT dynamics, and leverage it proactively to enhance CACC performance. Compared to passive schemes that simply acknowledge communication failures, a key innovation is to determine the optimal IFT by dynamically and proactively activating or deactivating communication devices of some CAVs in the platoon so as to mitigate negative effects of communication failures

and maximize string stability while factoring communication constraints.

(2). As another key innovation, the speed oscillation energy in frequency domain is used to evaluate the platoon control performance (i.e., string stability) for a given IFT degeneration scenario. This study treats the speed oscillation as a signal and determines the oscillation energy of each vehicle based on the transfer function of the given controller. The expected oscillation energy for an IFT is the weighted sum of the oscillation energies over all possible degeneration scenarios. Minimizing the expected speed oscillation energy implies maximizing the expected string stability.

(3). To account for the manifestation of different degeneration scenarios of the optimal IFT at different time instants within the time period, we design an adaptive control based on PD feedback controller and acceleration feedforward filter. When the IFT degenerates, the controller determines each vehicle's car-following behavior based on the information it receives at that time instant so that string stability is maintained.

To the best of our knowledge, this is the first study to use rigorous mathematical analysis to improve platoon performance by proactively leveraging IFT dynamics and adjusting adaptive controller parameters. It contributes to the literature in this area and informs the design of CAV platoon control in practice.

The remainder of the paper is organized as follows. Section 2 briefly introduces IFT and degeneration scenarios. Section 3 formulates the IFT optimization model. Section 4 formulates the adaptive controller for an IFT and its degeneration scenarios. Section 5 discusses several critical properties of the proposed CACC-OIFT strategy and discusses the solution algorithm for the IFT optimization problem. Section 6 discusses simulation-based numerical experiments and analyzes the results. Section 7 provides some concluding comments.

2. IFT and degeneration scenarios

Though the one-predecessor-following VIFT is the most commonly-used scheme, the more computationally intensive[1] two-predecessor-following VIFT is used in this study to illustrate the CACC-OIFT strategy. It should be noted that our model can be extended to a k-predecessor-following VIFT without loss of generality; however, computational efficiency and communication reliability issues can arise as more predecessors are considered. Fig. 2 shows a CAV platoon in which a fully-activated two-predecessor-following VIFT is used in the proposed adaptive PD controller. The information of each CAV is delivered to the two vehicles immediately following it through V2V communications. CAV i obtains the state of its two predecessors ($i-1$ and $i-2$), such as location (x_{i-1} and x_{i-2}), speed ($\dot{x}_{i-1}$ and $\dot{x}_{i-2}$) and acceleration ($\ddot{x}_{i-1}$ and $\ddot{x}_{i-2}$), through V2V communications. Also, vehicle i can detect the kinematic state (x_{i-1} and $\dot{x}_{i-1}$) of its immediate predecessor $i-1$ through onboard sensors such as radar, Lidar and camera, and its own kinematic state through GPS. The ambient traffic conditions, such as average density $\bar{k}$ and the trajectory oscillations in frequency domain[2] $X(j\omega)$, can be obtained through vehicle-to-infrastructure (V2I) communications, where ω is the angular frequency, and $j=\sqrt{-1}$.

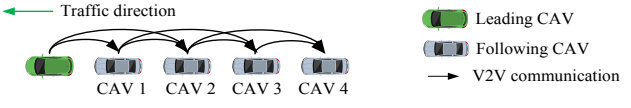


Fig. 2. CAV platoon with a two-predecessor-following VIFT scheme.

Since CACC-OIFT involves dynamically deactivating or activating the "send" functionality of V2V communication devices for vehicles in the platoon, we introduce a vector $\boldsymbol{\xi}=[\eta_0,\eta_1,\dots,\eta_N],\eta_i\in\{0,1\}$ for $i=0,1,\dots,N$ to indicate the IFT of a platoon with $N+1$ vehicles, where η_i indicates the status of the V2V communication device of vehicle i: $\eta_i=0$, when "send" functionality of V2V communication is deactivated; otherwise, $\eta_i=1$. For example, the IFT in Fig. 2 has $\boldsymbol{\xi}=[1,1,1,1,1]$. If some vehicles turn off their "send" functionality, such as vehicles 1, 3 and 4 in Fig. 3(a), the IFT has $\boldsymbol{\xi}=[1,0,1,0,0]$. We denote $\boldsymbol{\Omega}$ as the set of all possible IFTs that follow the two-predecessor-following scheme.

[1] In a VIFT, if the V2V device sends information to c other devices, it has 2^c communication statuses.

[2] The traffic oscillation in frequency domain measures oscillation amplitudes in different frequencies, which can be obtained through Fourier Transform of the ambient vehicles' trajectory data.



Fig. 3. Example of an IFT and its degeneration scenarios: (a) IFT with "send" functionalities of CAVs 1, 3 and 4 deactivated; (b) Degeneration scenario with CAV 2 failing to send message; (c) Degeneration scenario with CAV 0 failing to send message; (d) Degeneration scenario with both CAVs 0 and 2 failing to send messages.

Though temporarily switching off V2V communications of some vehicles can improve the success rate of other V2V communication links, communication failures cannot be eliminated as they also depend on ambient traffic conditions. As discussed earlier, we focus on failures involving the sending process. Due to sender failure, IFT $\boldsymbol{\xi}$ has degeneration scenarios $\boldsymbol{\xi_d}$ with probabilities $P_d(\boldsymbol{\xi_d})$, which can be formulated in a contention model of V2V communications (Qiu et al., 2015). Here, d is the index of degeneration scenarios, $d = 1, \dots D(\boldsymbol{\xi})$, where $D(\boldsymbol{\xi}) = 2^{\sum_{i=0}^{N} \eta_i}$. The degeneration scenario satisfies $\boldsymbol{\xi_d}(\boldsymbol{\xi}) = [\eta_{0,d}, \eta_{1,d}, \dots, \eta_{N,d}], \eta_{i,d} \in \{0,1\}, \eta_{i,d} \leq \eta_i$ for $i = 0,1, \dots, N$, which indicates that sender failure only exists for a vehicle with an activated "send" functionality. Hence, degeneration scenario $\boldsymbol{\xi_d}$ is related to IFT $\boldsymbol{\xi}$. For example, the IFT $\boldsymbol{\xi} = [1,0,1,0,0]$ in Fig. 3(a) has four degeneration scenarios: $\boldsymbol{\xi_1}(\boldsymbol{\xi}) = [1,0,1,0,0]$, $\boldsymbol{\xi_2}(\boldsymbol{\xi}) = [1,0,0,0,0]$, $\boldsymbol{\xi_3}(\boldsymbol{\xi}) = [0,0,1,0,0]$ and $\boldsymbol{\xi_4}(\boldsymbol{\xi}) = [0,0,0,0,0]$, which are shown in Fig. 3(a)-3(d), respectively. We denote $\boldsymbol{\Omega_d}(\boldsymbol{\xi})$ as the set of all possible degeneration scenarios $\boldsymbol{\xi_d}(\boldsymbol{\xi})$ for IFT $\boldsymbol{\xi}$.

To illustrate the need for controller design, we also analyze the V2V communication status from the receiver side. Based on different sender failures or deactivated "send" functionalities, a receiver (i.e., vehicle i in Fig. 4) has four possible communication statuses (Fig. 4(a)-4(d)). For a following vehicle i, if both predecessors (i.e., $i-1$ and $i-2$) activate communication devices and send information successfully, then the following vehicle i will be controlled by a CACC1 controller with the two-predecessor-following scheme in Fig. 4(a). Fig. 4(b) and 4(c) show the cases where one sender fails to broadcast its message. In these cases, CAV i can detect the kinematic state of its immediate predecessor $i-1$, and obtain one predecessor vehicle's acceleration through V2V communications. When both senders fail (Fig. 4 (d)), CAV i detects the surrounding environment only through onboard sensors. Then, the CACC will degrade to ACC to update the acceleration command based on the relative spacing and speed between CAVs i and $i-1$. Accordingly, a vector is introduced $\zeta(\boldsymbol{\xi_d}) = [\zeta_1(\boldsymbol{\xi_d}(\boldsymbol{\xi})), \dots, \zeta_N(\boldsymbol{\xi_d}(\boldsymbol{\xi}))], \zeta_i \in \{1,2,3,4\}$ for $i = 0,1, \dots, N$ in Equation (1) to indicate receiver status of a platoon with a degeneration scenario $\boldsymbol{\xi_d}$, where $\zeta_i(\boldsymbol{\xi_d}(\boldsymbol{\xi})) = 1,2,3,4$ indicates that vehicle i is controlled by CACC1, CACC2, CACC3, or ACC, respectively.

$$\zeta(\boldsymbol{\xi_d}(\boldsymbol{\xi})) = \begin{bmatrix} \zeta_0(\boldsymbol{\xi_d}(\boldsymbol{\xi})) \\ \zeta_1(\boldsymbol{\xi_d}(\boldsymbol{\xi})) \\ \vdots \\ \zeta_N(\boldsymbol{\xi_d}(\boldsymbol{\xi})) \end{bmatrix}^T = \begin{bmatrix} 4 \\ 4 \\ \vdots \\ 4 \end{bmatrix}^T - \boldsymbol{\xi_d}(\boldsymbol{\xi}) \begin{bmatrix} 0 & 2 & 1 & 0 & \cdots & 0 \\ \vdots & 0 & 2 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & 2 & 1 \\ \vdots & \ddots & \ddots & \ddots & 0 & 2 \\ 0 & \cdots & \cdots & \cdots & \cdots & 0 \end{bmatrix} \tag{1}$$

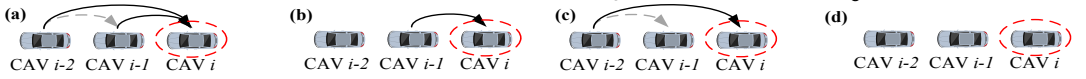


Fig. 4. Communication statuses of: (a) CACC1; (b) CACC2; (c) CACC3; (d) ACC.

3. Formulation of optimization model for the IFT

This section analytically derives the IFT optimization model for a time period. As discussed earlier, due to the dynamics of V2V communication failures, a platoon with IFT $\boldsymbol{\xi}$ may operate under different time-varying degeneration scenarios $\boldsymbol{\xi_d}(\boldsymbol{\xi})$ at different time instants with corresponding probabilities $P_d(\boldsymbol{\xi_d}(\boldsymbol{\xi}))$. The probabilities of degeneration scenarios of IFT $\boldsymbol{\xi}$ satisfy $\sum_{\boldsymbol{\xi_d}(\boldsymbol{\xi}) \in \boldsymbol{\Omega_d}(\boldsymbol{\xi})} P_d(\boldsymbol{\xi_d}(\boldsymbol{\xi})) = 1$. The platoon control performance is a function of the degeneration scenarios. Since degeneration scenario $\boldsymbol{\xi_d}$ is a function of IFT $\boldsymbol{\xi}$, the control performance under it is denoted by $E_d(\boldsymbol{\xi_d}(\boldsymbol{\xi}))$. By considering all possible degeneration scenarios $\boldsymbol{\Omega_d}(\boldsymbol{\xi})$, the expected platoon control performance $E(\boldsymbol{\xi})$ for IFT $\boldsymbol{\xi}$ under communication failures is:

$$E(\boldsymbol{\xi}) = \sum_{\boldsymbol{\xi_d}(\boldsymbol{\xi}) \in \boldsymbol{\Omega_d}(\boldsymbol{\xi})} P_d(\boldsymbol{\xi_d}(\boldsymbol{\xi})) E_d(\boldsymbol{\xi_d}(\boldsymbol{\xi})) \tag{2}$$

Hence, the choice of IFT $\boldsymbol{\xi}$ significantly affects the expected platoon performance, implying the need to determine the IFT that optimizes the expected performance $E(\boldsymbol{\xi})$ under the CACC strategy. Optimizing the IFT in terms of $E(\boldsymbol{\xi})$ aims to achieve tradeoff between the control performance under perfect communication conditions

and the robustness of the IFT in terms of communication reliability. We summarize the IFT optimization model, denoted as OPT-I, as follows:

$$
\begin{aligned}
\text{OPT-I} \quad & \underset{\boldsymbol{\xi}\in\boldsymbol{\Omega}}{\text{OPT}}\, E(\boldsymbol{\xi}) = \sum\nolimits_{\boldsymbol{\xi_d}(\boldsymbol{\xi})\in\boldsymbol{\Omega_d}(\boldsymbol{\xi})} P_d\big(\boldsymbol{\xi_d}(\boldsymbol{\xi})\big) E_{\boldsymbol{d}}\big(\boldsymbol{\xi_d}(\boldsymbol{\xi})\big) \\
\text{s.t.} \quad & \boldsymbol{\xi} = [\eta_0, \eta_1, \dots, \eta_N], \eta_i \in \{0,1\} \text{ for } i = 0,1,\dots,N \\
& \boldsymbol{\Omega} = \{[\eta_0, \eta_1, \dots, \eta_N] | \eta_i \in \{0,1\} \text{ for } i = 0,1,\dots,N \\
& \boldsymbol{\xi} \in \boldsymbol{\Omega} \\
& \boldsymbol{\xi_d}(\boldsymbol{\xi}) = \big[\eta_{0,d}, \eta_{1,d}, \dots, \eta_{N,d}\big], \eta_{i,d} \in \{0,1\}, \eta_{i,d} \le \eta_i \text{ for } i = 0,1,\dots,N \\
& \boldsymbol{\Omega_d}(\boldsymbol{\xi}) = \big\{\big[\eta_{0,d}, \eta_{1,d}, \dots, \eta_{N,d}\big] \,\big|\, \eta_{i,d} \in \{0,1\}, \eta_{i,d} \le \eta_i \text{ for } i = 0,1,\dots,N \\
& \textstyle\sum_{\boldsymbol{\xi_d}(\boldsymbol{\xi})\in\boldsymbol{\Omega_d}(\boldsymbol{\xi})} P_d(\boldsymbol{\xi_d}) = 1, \text{ for any } \boldsymbol{\xi} \in \boldsymbol{\Omega}
\end{aligned}
\tag{3}
$$

The first three constraints of OPT-I relate to the decision variable $\boldsymbol{\xi}$. The first constraint states that $\boldsymbol{\xi}$ is a binary 0-1 vector. The second constraint is the set $\boldsymbol{\Omega}$ of IFTs $\boldsymbol{\xi}$ corresponding to the two-predecessor-following VIFT. The third constraint states that $\boldsymbol{\xi}$ belongs to $\boldsymbol{\Omega}$. The remaining three constraints correspond to the degeneration scenario $\boldsymbol{\xi_d}(\boldsymbol{\xi})$. The fourth constraint shows the relationship between degeneration scenario $\boldsymbol{\xi_d}(\boldsymbol{\xi})$ and IFT $\boldsymbol{\xi}$. The fifth constraint indicates that the set $\boldsymbol{\Omega_d}(\boldsymbol{\xi})$ includes all possible degeneration scenarios for IFT $\boldsymbol{\xi}$. The last constraint states that the probabilities of the degeneration scenarios for an IFT $\boldsymbol{\xi}$ should sum up to 1.

Next, in Section 3.1, the platoon control performance is first characterized in terms of the speed oscillation energy of the vehicles in the platoon, which is then linked to string stability. Section 3.2 first discusses the determination of the probabilities of the degeneration scenarios. Then, it characterizes the expected string stability for an IFT $\boldsymbol{\xi}$ in terms of the speed oscillation energies of the platoon vehicles. Finally, the IFT optimization model is formulated in terms of optimizing the expected speed oscillation energies to reflect the expected string stability. Note that other performance metrics like energy consumption and cost can be included in this model using additional weighted terms in the objective function or by adding external constraints. Further, the focus on string stability does not affect safety considerations in the sense that channels (Qiu et al., 2015) used to propagate safety-related messages are different from those used for beacon messages (which are focus of this study).

3.1 Speed oscillation energy of a degeneration scenario for an IFT and its linkage to string stability

The traffic oscillations can be measured by the speed profiles of vehicles in the traffic flow (Li et al., 2012), which can be interpreted as a signal with specific energy density propagating through a certain medium (i.e., a platoon of vehicles). Correspondingly, this study quantifies traffic oscillations by introducing speed oscillation energy e_i in frequency domain for vehicle i:

$$e_i = \int_0^{+\infty} V_i^2(j\omega)\, d\omega \tag{4}$$

where $j = \sqrt{-1}$. $V_i^2(j\omega)$is the Energy Spectral Density (ESD) of vehicle speed. Thereby, the oscillation energy of a vehicle can be calculated by integrating its speed ESD over all frequencies. The speed oscillation energy of all vehicles in a platoon under a degeneration scenario $\boldsymbol{\xi_d}(\boldsymbol{\xi})$ is given by:

$$E_{\boldsymbol{d}}\big(\boldsymbol{\xi_d}(\boldsymbol{\xi})\big) = \sum\nolimits_{i=0}^{N} e_i = \sum\nolimits_{i=0}^{N} \left[\int_0^{+\infty} V_i^2(j\omega)\, d\omega\right] \tag{5}$$

Since a platoon with CACC is an interconnected system, the speed oscillation energy of each following vehicle will be determined by the speed oscillation energy of its predecessors and the characteristics of its CACC controller. The relationship between the speed oscillation energy of the leading vehicle 0 and any following vehicle i can be obtained by recursively linking the speed oscillation energy of the following vehicle and its predecessors. As discussed in Section 2, the trajectory oscillation information can be directly obtained through V2I communications, unlike speed oscillation information which is an indirect second-order effect. Hence, based on the CACC controller used in degeneration scenario $\boldsymbol{\xi_d}(\boldsymbol{\xi})$, a transfer function $\mathrm{SS}_{X,i}(j\omega, \boldsymbol{\xi_d}(\boldsymbol{\xi}))$ is introduced to measure the propagation of trajectory oscillations in frequency domain from the leading vehicle 0 to any following vehicle i:

$$\mathrm{SS}_{X,i}\big(j\omega, \boldsymbol{\xi_d}(\boldsymbol{\xi})\big) = \frac{X_i(j\omega)}{X_0(j\omega)}, \quad i = 1, \dots, n \tag{6}$$

where $X_i(j\omega)$ is the position frequency response of vehicle i. Next, we assume that the position frequency response of the leading vehicle $X_0(j\omega)$ follows the ambient traffic oscillations in frequency domain $X(j\omega)$ (i.e., $X_0(j\omega) =$

$X(j\omega)$), since the movement of the leading vehicle is affected by the ambient traffic oscillations.

Given the degeneration scenario $\boldsymbol{\xi_d(\xi)}$, $\mathrm{SS}_{X,i}(j\omega, \boldsymbol{\xi_d(\xi)})$ is a function of ω with a set of predetermined controller parameters. Different transfer functions will be generated under different controllers and IFT degeneration scenarios. Section 4 provide more details on $\mathrm{SS}_{X,i}(j\omega, \boldsymbol{\xi_d(\xi)})$.

Next, we derive the speed frequency response $V_i(j\omega)$ for all vehicles using the leading vehicle position frequency response $X_0(j\omega)$ information. To do so, first, the inverse Fourier transform is performed to obtain the trajectory information of vehicle i in time domain.

$$x_i(t) = \int_0^{+\infty} X_i(j\omega) e^{2\pi t j\omega} d\omega = \int_0^{+\infty} \mathrm{SS}_{X,i}\big(j\omega, \boldsymbol{\xi_d(\xi)}\big) X_0(j\omega) e^{2\pi t j\omega} d\omega \tag{7}$$

Then, the derivative of the vehicle trajectory provides the speed profile of vehicle i in time domain.

$$v_i(t) = \frac{dx_i(t)}{dt} = \int_0^{+\infty} 2\pi j\omega \mathrm{SS}_{X,i}\big(j\omega, \boldsymbol{\xi_d(\xi)}\big) X_0(j\omega) e^{2\pi t j\omega} d\omega \tag{8}$$

Comparing Equation (8) with the inverse Fourier transform of speed frequency response in Equation (9),

$$v_i(t) = \int_0^{+\infty} V_i(j\omega) e^{2\pi t j\omega} d\omega \tag{9}$$

the speed frequency response is derived as follows:

$$V_i(j\omega) = 2\pi j\omega \mathrm{SS}_{X,i}\big(j\omega, \boldsymbol{\xi_d(\xi)}\big) X_0(j\omega) \tag{10}$$

Combining Equations (5) and (10), the speed oscillation energy of the vehicles in the platoon under a degeneration scenario $\boldsymbol{\xi_d(\xi)}$ can be formulated as:

$$E_d\big(\boldsymbol{\xi_d(\xi)}\big) = \sum_{i=0}^{N} \int_0^{+\infty} V_i^2(j\omega) d\omega = 4\pi^2 \sum_{i=0}^{N} \int_0^{+\infty} \omega^2 \mathrm{SS}_{X,i}^2\big(j\omega, \boldsymbol{\xi_d(\xi)}\big) X_0^2(j\omega) d\omega \tag{11}$$

The transfer function $\mathrm{SS}_{X,i}\big(j\omega, \boldsymbol{\xi_d(\xi)}\big)$ is commonly used to infer string stability (Naus et al., 2010). A lower value for $\mathrm{SS}_{X,i}\big(j\omega, \boldsymbol{\xi_d(\xi)}\big)$ reflects better string stability performance as it implies smoother traffic flow. From Equation (11), for a given leading vehicle position frequency response $X_0(j\omega)$, a smaller platoon speed oscillation energy $E_d\big(\boldsymbol{\xi_d(\xi)}\big)$ indicates a lower $\mathrm{SS}_{X,i}\big(j\omega, \boldsymbol{\xi_d(\xi)}\big)$, implying better string stability performance.

3.2 Probabilities of degeneration scenarios and expected string stability for an IFT

Since the optimal IFT implies that some vehicles have their "send" functionality activated and others have it deactivated, a communication failure implies that an activated "sender" vehicle fails to broadcast its message at a specific time instant. This is a manifestation of a degeneration scenario for this IFT. As discussed in Section 1, sender failure relates to information collision during message broadcasting in the DSRC protocol. This collision occurs when two senders within communication range send information to a receiver at the same time. During the sending process, each sender uses a contention window (CW) to compete for a sending chance. Each sender will randomly select an integer value in the range of $[0, \mathrm{CW}-1]$, to determine when in this window it will send the message. If more than one sender chooses the same integer value, a collision occurs. Note that we use the DSRC protocol here as an example of the communication protocol to illustrate our CACC-OIFT strategy because it is a mature and proven technology that has been invested in by both the public and private sectors. However, the proposed strategy can be used with any communication protocol with capacity upper bound where the communication success rate depends on the density of vehicles with communication ability. Hence, it is always possible to improve platoon performance by optimizing IFT as in the future more information (traffic, entertainment, weather, e-commerce, etc.) will sought to be propagated or shared among vehicles, even if high-capacity protocols such as 5G are ultimately adopted.

In the MAC level protocol (i.e., IEEE 802.11), collision probability can be reduced by increasing the contention window CW size and/or implementing a retransmission scheme. However, both these methods increase time delay, reducing the timeliness of information propagation. For example, if a collision happens, the sender will choose a new contention window and attempts to retransmit the same information after the previous contention window ends (Hafeez et al. 2013). This will lead to delays in the information reaching the receiver, worsening the performance of the real-time controller. Hence, these two methods are not suitable for CACC. Further, as information is generated continuously in the CAV context, if a sender fails to win the sending chance before the next information is generated, the previous information will be dropped, and will therefore also be identified as a sender failure (Qiu et al. 2015). Therefore, a retransmission scheme is not considered here and the contention window CW size is set to a small value.

Also, usually communication time delay is not large compared to the control sampling period and will not significantly impact platoon performance. Based on these assumptions, we do not consider communication delay here.

In Qiu et al. (2015), a contention model with saturated and unsaturated communication traffic is developed using a Markov chain. The success rate of sending message for vehicle i is:

$$p_{i,\text{sat}} = 2(1-b_i)(1-2b_i+\text{CW})^{-1} \tag{12}$$

where the channel busy rate b_i for sender vehicle i is:

$$b_i = 1 - e^{-\bar{\rho}_i(\boldsymbol{\xi})p_{i,\text{sat}}} \tag{13}$$

Here, $\bar{\rho}_i(\boldsymbol{\xi})$ is the average number of vehicles with activated "send" functionalities within communication range R of vehicle i. Given the average density ($\bar{k}$) of the ambient traffic flow, the average number of vehicles within the communication range R is given by $m = \lfloor R\bar{k} \rfloor$. Then, the vector of $\bar{\rho}(\boldsymbol{\xi}) = [\bar{\rho}_0(\boldsymbol{\xi}), \bar{\rho}_1(\boldsymbol{\xi}), \dots, \bar{\rho}_N(\boldsymbol{\xi})]$ is:

$$\bar{\rho}(\boldsymbol{\xi}) = \boldsymbol{\xi}\boldsymbol{M}(\bar{k}) \tag{14}$$

where, $\boldsymbol{M}(\bar{k})$ is a N-by-N $2m+1$ diagonal matrix whose non-zero elements are 1 if $m < N$. Otherwise, M is a N-by-N matrix in which all elements have the value 1.

Combining Equations (12) and (13), the success rate of sending, $p_{i,\text{sat}}$, can be solved numerically based on a method by Qiu et al. (2012), and for an IFT $\boldsymbol{\xi}$, is denoted as $p_{i,sat}(\boldsymbol{\xi})$.

Since the V2V device bandwidth and its impact are not considered, a contention model with unsaturated communication is implemented (Qiu et al., 2015). The success rate $p_{i,unsat}$ of a sender vehicle i in one attempt is:

$$p_{i,unsat}(\boldsymbol{\xi}) = [k_1 \log(\bar{\rho}_i(\boldsymbol{\xi})) + k_2\text{CW} + k_3]p_{i,sat}(\boldsymbol{\xi}) \tag{15}$$

where k_1, k_2, k_3 are fitting coefficients. In our numerical experiments, they are fitted using the simulation results in NS-3 through linear regression.

After obtaining the success rate of a sender vehicle, the probabilities of the degeneration scenarios of an IFT due to sender failure can be calculated. When the IFT degenerates from $\boldsymbol{\xi}$ to $\boldsymbol{\xi_d}$, two sets $\boldsymbol{A_d}(\boldsymbol{\xi_d}(\boldsymbol{\xi}))$ and $\boldsymbol{B_d}(\boldsymbol{\xi_d}(\boldsymbol{\xi}))$ are introduced for the indices of vehicles with successful and unsuccessful broadcast status, respectively, which satisfy $\boldsymbol{A_d}(\boldsymbol{\xi_d}(\boldsymbol{\xi})) = \{i|\eta_i = 1, \eta_{i,d} = 1, i = 0, \dots, N\}$ and $\boldsymbol{B_d}(\boldsymbol{\xi_d}(\boldsymbol{\xi})) = \{i|\eta_i = 1, \eta_{i,d} = 0, i = 0, \dots, N\}$. Hence, the probability of the degeneration from $\boldsymbol{\xi}$ to $\boldsymbol{\xi}_d$ is:

$$P_d(\boldsymbol{\xi_d}(\boldsymbol{\xi})) = \prod\nolimits_{i\in\boldsymbol{A_d}(\boldsymbol{\xi})} p_{i,\text{unsat}} \prod\nolimits_{i\in\boldsymbol{B_d}(\boldsymbol{\xi})} (1-p_{i,\text{unsat}}) \tag{16}$$

The probability of each degeneration scenario is independent of its string stability performance. By substituting Equations (11) and (16) into OPT-I, the optimization model can be reformulated as OPT-II.

OPT-II

$$\min_{\boldsymbol{\xi}\in\boldsymbol{\Omega}} 4\pi^2 \sum_{\boldsymbol{\xi}_d(\boldsymbol{\xi})\in\boldsymbol{\Omega}_d(\boldsymbol{\xi})} \left[\prod_{i\in\boldsymbol{A_d}(\boldsymbol{\xi})} p_{i,\text{unsat}} \sum_{i\in\boldsymbol{B_d}(\boldsymbol{\xi})} (1-p_{i,\text{unsat}}) \sum_{i=1}^{N} \int_0^{+\infty} \omega^2 \text{SS}_{X,i}^2(j\omega, \boldsymbol{\xi_d}(\boldsymbol{\xi})) X_0^2(j\omega) d\omega \right] \tag{17}$$

s.t.

$$\boldsymbol{\xi} = [\eta_0, \eta_1, \dots, \eta_N], \eta_i \in \{0,1\} \text{ for } i = 0,1,\dots,N$$

$$\boldsymbol{\Omega} = \{[\eta_0, \eta_1, \dots, \eta_N] | \eta_i \in \{0,1\} \text{ for } i = 0,1,\dots,N\}$$

$$\boldsymbol{\xi} \in \boldsymbol{\Omega}$$

$$\boldsymbol{\xi_d}(\boldsymbol{\xi}) = [\eta_{0,d}, \eta_{1,d}, \dots, \eta_{N,d}], \eta_{i,d} \in \{0,1\}, \eta_{i,d} \le \eta_i \text{ for } i = 0,1,\dots,N$$

$$\boldsymbol{\Omega_d}(\boldsymbol{\xi}) = \{[\eta_{0,d}, \eta_{1,d}, \dots, \eta_{N,d}] \mid \eta_{i,d} \in \{0,1\}, \eta_{i,d} \le \eta_i \text{ for } i = 0,1,\dots,N\}$$

$$\textstyle\sum_{\xi_d(\xi)\in\boldsymbol{\Omega}_d(\xi)} P_d(\boldsymbol{\xi_d}) = 1, \text{ for any } \boldsymbol{\xi} \in \boldsymbol{\Omega}$$

$$\boldsymbol{A_d}(\boldsymbol{\xi_d}(\boldsymbol{\xi})) = \{i|\eta_i = 1, \eta_{i,d} = 1, i = 0, \dots, N\}$$

$$\boldsymbol{B_d}(\boldsymbol{\xi_d}(\boldsymbol{\xi})) = \{i|\eta_i = 1, \eta_{i,d} = 0, i = 0, \dots, N\}$$

OPT-II is a mixed binary-integer optimization problem. It is almost identical to OPT-I, but has two additional constraints at the end related to the successful and unsuccessful broadcast status under the degeneration scenario $\boldsymbol{\xi_d}(\boldsymbol{\xi})$ for IFT $\boldsymbol{\xi}$, respectively. The objective function seeks the IFT whose most likely degeneration scenarios have lower speed oscillation energies (or, higher string stability performance). Section 5 proposes an algorithm to solve OPT-II and discusses its characteristics.

4. Formulation of adaptive controller for degeneration scenarios of IFT

We propose an adaptive Proportional-Derivative (PD) controller based on a two-predecessor-following scheme. First, we introduce the control structure (including vehicle dynamics, spacing policy, and feedforward and feedback sub-controllers) that can adapt to any IFT and its degeneration scenarios. Next, we determine several critical

parameters to ensure the head-to-tail string stability of the platoon and improve the capability for measurement noise mitigation for individual vehicles. The transfer function $SS_{X,i}(j\omega, \boldsymbol{\xi_d}(\boldsymbol{\xi}))$ is used in the IFT optimization model as the indicator of string stability performance for all degeneration scenarios $\boldsymbol{\xi_d}(\boldsymbol{\xi})$ of IFT $\boldsymbol{\xi}$.

4.1 Control structure

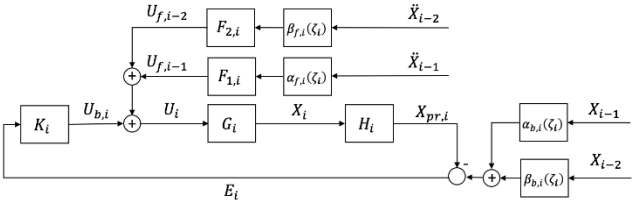


Fig. 5. Schematic of the adaptive PD controller.

The schematic of the adaptive PD controller for vehicle i is illustrated in Fig. 5. U_i represents the control command, which consists of control feedback $U_{b,i}$ from the spacing error E_i and two extra feedforward terms $U_{f,i-1}$ and $U_{f,i-2}$ from the acceleration rates $\ddot{X}_{i-1}$ and $\ddot{X}_{i-2}$, respectively. In the case of ACC, U_i merely consists of a feedback control command. X_i is the position output of vehicle i, $X_{pr,i}$ is the processed position output of vehicle i after considering the spacing policy, X_{i-1} is the feedback position information from the immediate predecessor while X_{i-2} is the feedback position information from the second predecessor. K_i is the feedback controller which generates a control command to rectify the spacing error. G_i represents the ideal longitudinal vehicle dynamics. H_i denotes the spacing policy (i.e., such as CD or CTH), and $F_{1,i}$ and $F_{2,i}$ are feedforward filters to process the acceleration information from the corresponding predecessors. $\alpha_{b,i}(\zeta_i)$ and $\beta_{b,i}(\zeta_i)$ are weighting coefficients for position feedback information while $\alpha_{f,i}(\zeta_i)$ and $\beta_{f,i}(\zeta_i)$ are weighting coefficients for acceleration feedforward information. These coefficients are determined by the dynamic IFT; specific coefficient settings for each IFT scenario are shown in Table 1.

Table 1. Weighting coefficient settings.

	ζ_i	$\alpha_{b,i}(\zeta_i)$	$\alpha_{f,i}(\zeta_i)$	$\beta_{b,i}(\zeta_i)$	$\beta_{f,i}(\zeta_i)$
CACC1	1	α	α	β	β
CACC2	2	1	1	0	0
CACC3	3	1	0	0	1
ACC	4	1	0	0	0

The extra constraints for CACC1 are:

$$\alpha + \beta = 1,\ 0 < \alpha, \beta < 1 \tag{18}$$

CACC1 has two predecessors with activated "send" functionality; thereby, the weighting coefficients of feedback and feedforward information are both greater than zero. The weighting coefficients α and β represent the relative importance of the information from the immediate and second predecessors, respectively. As CACC2 only has the immediate predecessor with activated "send" functionality, the weighting coefficient of the information from the second predecessor is zero. Similarly, the weighting coefficient of the information from the immediate predecessor in CACC3 is zero, and both weighting coefficients in ACC case are zero.

4.3.1. Vehicle dynamics

This study considers an idealized longitudinal vehicle dynamics model, ignoring air drag, rolling resistance and actuator delay (Li et al., 2015) in the model. This model is selected because the study focuses on upper-level control design (damping traffic oscillations), and assumes that the lower-level controller can address vehicle internal dynamics so that the vehicle can respond to acceleration commands without delays. Additionally, it is common to use such an idealized model for system-level platoon control design (Gong et al., 2016; Zhou et al., 2017). The linearized state-space representation of the idealized longitudinal vehicle dynamics can be represented as:

$$\dot{x}_i(t) = v_i(t), \qquad \dot{v}_i(t) = u_i(t) \tag{19}$$

where $x_i(t)$, $v_i(t)$, and $u_i(t)$ are the absolute position, velocity and acceleration of vehicle i at time t, respectively.

To analyze stability performance, the modeling and analysis are performed in the Laplace domain. The idealized

longitudinal vehicle dynamics model in the Laplace domain can be described by using a transfer function:

$$G_i(s) = X_i(s)U_i(s)^{-1} = s^{-2} \tag{20}$$

where the input $U_i(s)$ denotes the acceleration of vehicle i and the output $X_i(s)$ denotes the absolute position of vehicle i in the Laplace domain.

4.3.2. Spacing policy

To achieve more efficient damping oscillations, we obtain the desired relative distances between vehicle i and its two predecessors using the CTH policy as follows:

$$d_{i,1}(t) = L + h\dot{x}_i(t), \qquad d_{i,2}(t) = 2[L + h\dot{x}_i(t)] \tag{21}$$

where $d_{1,i}(t)$ is the desired relative distance between vehicle i and vehicle $i-1$, and $d_{2,i}(t)$ is the desired relative distance between vehicle i and vehicle $i-2$. L is the constant standstill distance (including vehicle length) between the two vehicles, $\dot{x}_i(t)$ is the velocity of vehicle i, and h is the desired time headway.

The convex combination of spacing errors in Equation (22) is implemented in the feedback loop since the feedback controller processes spacing errors from both predecessors.

$$\begin{aligned} e_i(t) &= \alpha_{b,i}(\zeta_i)\{x_{i-1}(t) - x_{pr,i,1}(t)\} + \beta_{b,i}(\zeta_i)\{x_{i-2}(t) - x_{pr,i,2}(t)\} \\ &= \alpha_{b,i}(\zeta_i)\{[x_{i-1}(t) - x_i(t)] - d_{i,1}(t)\} + \beta_{b,i}(\zeta_i)\{[x_{i-2}(t) - x_i(t)] - d_{i,2}(t)\} \end{aligned} \tag{22}$$

From Table 1, we note that $\alpha_{b,i}(\zeta_i) + \beta_{b,i}(\zeta_i) = 1$ is always satisfied in the four controller statuses. Substituting Equation (21) into Equation (22), the weighted spacing error is

$$e_i(t) = \alpha_{b,i}(\zeta_i)x_{i-1}(t) + \beta_{b,i}(\zeta_i)x_{i-2}(t) - x_i(t) - (2 - \alpha_{b,i}(\zeta_i))(L + h\dot{x}_i(t)) \tag{23}$$

Taking the Laplace transformation of Equation (23), the spacing error can be expressed equivalently as:

$$E_i(s) = \alpha_{b,i}(\zeta_i)X_{i-1}(s) + \beta_{b,i}(\zeta_i)X_{i-2}(s) - H_i(s)X_i(s) \tag{24}$$

where $H_i(s)$ is the CTH spacing policy in frequency domain, given by:

$$H_i(s) = 1 + \left(2 - \alpha_{b,i}(\zeta_i)\right)hs \tag{25}$$

4.3.3. Acceleration feedforward

From Fig. 5, the relationship between tracking error $E_i(s)$ and the feedforward acceleration $\ddot{X}_{i-1}(s) = s^2X_{i-1}(s)$ of the predecessor $i-1$ and feedforward acceleration $\ddot{X}_{i-2}(s) = s^2X_{i-2}(s)$ of the predecessor $i-2$ in the Laplace domain is formulated as:

$$E_i(s) = \frac{\alpha_{b,i}(\zeta_i) - \alpha_{f,i}(\zeta_i)H_i(s)G_i(s)F_{1,i}(s)s^2}{1 + H_i(s)G_i(s)K_i(s)}X_{i-1}(s) + \frac{\beta_{b,i}(\zeta_i) - \beta_{f,i}(\zeta_i)H_i(s)G_i(s)F_{2,i}(s)s^2}{1 + H_i(s)G_i(s)K_i(s)}X_{i-2}(s) \tag{26}$$

To eliminate spacing error between adjacent vehicles, feedforward filters $F_{1,i}(s)$ and $F_{2,i}(s)$ are designed based on a zero-error condition (Naus et al., 2010). Hence, the numerators of the first and second terms on the right hand side in Equation (26) should be zero. Doing so, and substituting the $\alpha_{b,i}(\zeta_i)$, $\alpha_{f,i}(\zeta_i)$, $\beta_{b,i}(\zeta_i)$ and $\beta_{f,i}(\zeta_i)$ from Table 1 and the spacing policy $H_i(s)$ from Equation (25) into it, we can derive the feedforward filters as:

$$F_{1,i}(s) = F_{2,i}(s) = (H_i(s)G_i(s)s^2)^{-1} = \left(H_i(s)\right)^{-1} \tag{27}$$

4.3.4. Control command

As illustrated in Fig. 5, our control command consists of a feedback term and two feedforward terms:

$$U_i(s) = U_{b,i}(s) + U_{f,i-1}(s) + U_{f,i-2}(s) \tag{28}$$

Recall that the feedback term $U_{b,i}(s)$ uses spacing error to stabilize the closed-loop system while the feedforward terms $U_{f,i-1}(s)$ and $U_{f,i-2}(s)$ use acceleration rates from the two predecessors to minimize the spacing error.

The feedback term $U_{b,i}(s)$ and the corresponding PD feedback controller are defined as:

$$U_{b,i}(s) = K_i(s)E_i(s) \tag{29}$$

$$K_i(s) = \omega_{K,i}\left(\omega_{K,i} + s\right) \tag{30}$$

where $E_i(s)$ is the spacing error in the Laplace domain in Equation (24). $\omega_{K,i}$ is the cut-off frequency of the PD controller. Cut-off frequency influences how aggressively vehicles respond to rectify spacing errors; a larger cut-off frequency will make vehicle react faster. More importantly, $\omega_{K,i}$ affects the value of string stability $\mathrm{SS}_{X,i}\left(j\omega, \boldsymbol{\xi_d}(\boldsymbol{\xi})\right)$ in Equation (6), and will be determined analytically in Section 4.2.

The feedforward terms $U_{f,i-1}(s)$ and $U_{f,i-2}(s)$ indicate that the acceleration rates of vehicles $i-1$ and $i-2$ are sent to i, respectively:

$$U_{f,i-1}(s) = \alpha_{f,i}(\zeta_i)F_{1,i}(s)s^2X_{i-1}(s) \tag{31}$$

$$U_{f,i-2}(s) = \beta_{f,i}(\zeta_i)F_{2,i}(s)s^2X_{i-2}(s) \tag{32}$$

Note that according to the two-predecessor-following scheme, the second vehicle in the platoon can receive acceleration information from only the leading vehicle, that is, the feedforward term of vehicle 1 includes only $U_{f,0}(s)$.

The overall control command is obtained by summing up Equations (29), (31) and (32). Through inverse Laplace transformation, the expression for the control command is:

$$u_i(t) = \omega_{K,i}^2 e_i(t) + \omega_{K,i}\dot{e}_i(t) + \alpha_{f,i}(\zeta_i)F_{1,i}(t)\ddot{x}_{i-1}(t) + \beta_{f,i}(\zeta_i)F_{2,i}(t)\ddot{x}_{i-2}(t) \quad (33)$$

The discretized version of (33) is used for operational deployment. We do not discuss it here due to space constraints.

4.2. Stability analysis and parameter determination

Two parameters in the system impact platoon performance: time headway h and cut-off frequency $\omega_{K,i}$. We analyze these parameters to mitigate measurement noise and reduce platoon oscillations to aid string stability.

4.2.1. Measurement noise mitigation

Measurement noise is a high-frequency noise generated from onboard sensors that produces inaccurate trajectory information, causing undesirable control inputs or platoon oscillations. Hence, mitigation of the measurement noise effect can improve control performance in terms of stability for individual vehicles in the platoon. Lemma 1 presents the characteristics of an upper bound for the product of h and $\omega_{K,i}$ (i.e., $h\omega_{K,i}$) to mitigate measurement noise. Additionally, for measurements with greater noise contamination, the settings of weighting coefficients in CACC1 can be adjusted to lower values to improve the noise mitigation effect.

Lemma 1: By setting an upper bound for the product of h and $\omega_{K,i}$ as: $h\omega_{K,i} \le W_{max}$, the high-frequency measurement noise from the two predecessors is individually attenuated by at least $W_{max}/[1 + W_{max}]$.

Proof: For any following vehicle i in the platoon, the source of the measurement noise is mainly from the movement state detection of the two predecessors $i-1$ and $i-2$. The measured position X_{i-1} (X_{i-2}) of predecessor $i-1$ ($i-2$) consists of true value of position $\bar{X}_{i-1}$ ($\bar{X}_{i-2}$) and measurement noise N_{i-1} (N_{i-2}): $X_{i-1} = \bar{X}_{i-1} + N_{i-1}$ ($X_{i-2} = \bar{X}_{i-2} + N_{i-2}$). From Fig. 5, the complementary sensitivity functions $T_{1,i}$ ($T_{2,i}$) can be used to describe the relationship between the processed position output $X_{pr,i}(s)$ of vehicle i and measurement noise N_{i-1} (N_{i-2}) included in position of predecessor $i-1$ ($i-2$).

$$T_{1,i}(s) = X_{pr,i}(s)N_{i-1}(s)^{-1} = \alpha_{b,i}(\zeta_i)H_i(s)G_i(s)K_i(s)[1 + H_i(s)G_i(s)K_i(s)]^{-1} \quad (34)$$

$$T_{2,i}(s) = X_{pr,i}(s)N_{i-2}(s)^{-1} = \beta_{b,i}(\zeta_i)H_i(s)G_i(s)K_i(s)[1 + H_i(s)G_i(s)K_i(s)]^{-1} \quad (35)$$

The magnitude of complementary sensitivity function $T_{1,i}$ ($T_{2,i}$) at a high-frequency represents the effect of measurement noise mitigation (a larger value of $T_{1,i}$ or $T_{2,i}$ indicates reduced mitigation of measurement noise). The key aspect of mitigating high-frequency measurement noise of X_{i-1} (X_{i-2}) is to decrease the value of $T_{1,i}$ ($T_{2,i}$) in the high-frequency band. Substituting $G_i(s)$, $H_i(s)$, and $K_i(s)$ from Equations (20), (25), and (30) into Equations (34)-(35), we have:

$$\lim_{s\to\infty} T_{1,i}(s) = \lim_{s\to\infty} \alpha_{b,i}(\zeta_i) \frac{h\omega_{K,i}s^2 + \left(h\omega_{K,i}{}^2 + \omega_{K,i}\right)s + \omega_{K,i}{}^2}{\left(1 + h\omega_{K,i}\right)s^2 + \left(h\omega_{K,i}{}^2 + \omega_{K,i}\right)s + \omega_{K,i}{}^2} = \alpha_{b,i}(\zeta_i)\frac{h\omega_{K,i}}{1 + h\omega_{K,i}} \quad (36)$$

$$\lim_{s\to\infty} T_{2,i}(s) = \lim_{s\to\infty} \beta_{b,i}(\zeta_i) \frac{h\omega_{K,i}s^2 + \left(h\omega_{K,i}{}^2 + \omega_{K,i}\right)s + \omega_{K,i}{}^2}{\left(1 + h\omega_{K,i}\right)s^2 + \left(h\omega_{K,i}{}^2 + \omega_{K,i}\right)s + \omega_{K,i}{}^2} = \beta_{b,i}(\zeta_i)\frac{h\omega_{K,i}}{1 + h\omega_{K,i}} \quad (37)$$

From Table 1, we have $\alpha_{b,i}(\zeta_i) \le 1$ and $\beta_{b,i}(\zeta_i) \le 1$. By setting an upper bound for the product of $h\omega_{K,i}$ as: $h\omega_{K,i} \le W_{max}$, the upper bounds of $\lim_{s\to\infty} T_{1,i}(s)$ and $\lim_{s\to\infty} T_{2,i}(s)$ can be determined as $\lim_{s\to\infty} T_{1,i}(s) \le h\omega_{K,i}/\left[1 + h\omega_{K,i}\right] \le W_{max}/[1 + W_{max}]$ and $\lim_{s\to\infty} T_{2,i}(s) \le h\omega_{K,i}/\left[1 + h\omega_{K,i}\right] \le W_{max}/[1 + W_{max}]$, respectively, which indicates that the high-frequency measurement noise from two predecessors is individually attenuated by at least $W_{max}/[1 + W_{max}]$. This completes the proof for Lemma 1.

For operational deployment, W_{max} depends on the measurement noise mitigation needs of the specific system. Its value is determined through trial-and-error. In the study experiments, we aim to attenuate the measurement noise by a factor of at least 2/3. By calculating $W_{max}/[1 + W_{max}] = 2/3$, the upper bound of $h_d\omega_{K,i}$ is set as $W_{max} = 2$.

Remark 1: In CACC1, the values of $\alpha_{b,i}(\zeta_i)$ and $\beta_{b,i}(\zeta_i)$ can be adjusted based on the signal-to-noise ratio (SNR)[3] of position measurement X_{i-1} and X_{i-2}: $\alpha_{b,i}(\zeta_i)/\beta_{b,i}(\zeta_i) = \mathrm{SNR2}/\mathrm{SNR1}$, where SNR1 corresponds to

[3]Signal-to-noise ratio (SNR) is defined as the ratio of the power of a signal divided by the power of measurement noise, which can also be expressed as the ratio between variance of measured signal and measurement noise: $\mathrm{SNR} = P_{signal}/P_{noise} = \sigma_{signal}^2/\sigma_{noise}^2$; a larger SNR indicates a weaker noise effect on the measurement information.

measurement of X_{i-1} and SNR2 corresponds to measurement of X_{i-2}. The adjustment of weighting coefficients can improve the effect of noise mitigation by putting lesser weight on measurements with smaller SNR.

4.2.2. String stability analysis

The string stability transfer function is specified as a measure of signal amplification. The platoon head-to-tail string stability is analyzed to ensure that traffic oscillations are effectively damped when they reach the platoon tail. A homogeneous string stability criterion is used as we assume a homogeneous vehicle platoon. The transfer function is the ratio of the trajectory oscillations in the Laplace domain of the ith vehicle to that of the leading vehicle:

$$\mathrm{SS}_{X,i}(s) = X_i(s)/X_0(s) \tag{38}$$

To ensure head-to-tail string stability, with $s = j\omega$, we have the string stability condition:

$$\left\|\mathrm{SS}_{X,i}(j\omega)\right\|_\infty = \left\|X_i(j\omega)/X_0(j\omega)\right\|_\infty \le 1 \tag{39}$$

where $j = \sqrt{-1}$, and the string stability condition based on ∞-norm can be satisfied by the requirement of $\left|\mathrm{SS}_{X,i}(j\omega)\right| \le 1$ for all ω. For simplification, in the following analysis, we use $\mathrm{SS}_{X,i}$, X_i, G_i, H_i, K_i, $F_{1,i}$, and $F_{2,i}$ to denote $\mathrm{SS}_{X,i}(s)$, $X_i(s)$, $G_i(s)$, $H_i(s)$, $K_i(s)$, $F_{1,i}(s)$, and $F_{2,i}(s)$, respectively.

From Equations (1), (20), and (25)-(32), the transfer functions of string stability of all vehicles in platoon are described as:

$$\overline{\mathrm{SS}} = T(\zeta)S \tag{40}$$

where $\overline{\mathrm{SS}} = \left[\mathrm{SS}_{\mathrm{x},0}\ \mathrm{SS}_{\mathrm{x},1}\ \mathrm{SS}_{\mathrm{x},2}\ \mathrm{SS}_{\mathrm{x},3}\ \cdots\ \mathrm{SS}_{x,n}\right]^T$, $S = [1\ 0\ 0\ 0\ \cdots\ 0]^T$, $\mathrm{SS}_{\mathrm{x},0} = 1$ and $\mathrm{SS}_{\mathrm{x,i}} = X_i/X_0$, $i \ge 1$.

$$T(\zeta) = \begin{bmatrix} 1 & 0 & 0 & 0 & \cdots & 0 \\ \mathcal{G}_{1,1} & -1 & 0 & 0 & \ddots & \vdots \\ \mathcal{G}_{2,2} & \mathcal{G}_{2,1} & -1 & 0 & \ddots & 0 \\ 0 & \mathcal{G}_{3,2} & \mathcal{G}_{3,1} & -1 & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & \ddots & 0 \\ 0 & \cdots & \cdots & \mathcal{G}_{n,2} & \mathcal{G}_{n,1} & -1 \end{bmatrix}^{-1}$$

$$\mathcal{G}_{i,1} = \alpha_{f,i}(\zeta_i)\Lambda_{f,i-1} + \alpha_{b,i}(\zeta_i)\Lambda_{b,i-1} \tag{41}$$

$$\mathcal{G}_{i,2} = \beta_{f,i}(\zeta_i)\Lambda_{f,i-2} + \beta_{b,i}(\zeta_i)\Lambda_{b,i-2} \tag{42}$$

where $\Lambda_{f,i-2} = \frac{G_i F_{2,i} s^2}{1+G_i K_i H_i}$is the transfer function between the position of vehicles $i-2$ and i with respect to the feedforward term $U_{f,i-2}(s)$; $\Lambda_{f,i-1} = \frac{G_i F_{1,i} s^2}{1+G_i K_i H_i}$ is the transfer function between vehicles $i-1$ and i with respect to feedforward term $U_{f,i-1}(s)$; $\Lambda_{b,i-1} = \Lambda_{b,i-2} = \frac{G_i K_i}{1+G_i K_i H_i}$ is the transfer function between the position of vehicles $i-1$ or $i-2$ and i with respect to the feedback term $U_{b,i}(s)$.

Based on the four communication statuses described in Fig. 4, we analyze the corresponding feasible regions for the time headway h and cut-off frequency $\omega_{K,i}$ to ensure string stability, using Proposition 1 and Lemma 2.

Proposition 1: If $\sigma \ge 0$, a first order transfer function: $q(j\omega) = \frac{1}{1+j\omega\sigma}$ satisfies the string stability in Equation (39).

Proof: Since $|q(j\omega)| = \frac{1}{\sqrt{1+\sigma^2\omega^2}}$ and phase angle $\angle q(j\omega) = -\arctan(\sigma\omega)$, the magnitude of $q(j\omega)$ will always be lesser than or equal to one. However, when $\sigma < 0$, the phase angle of $q(j\omega)$ is positive, which indicates that the system corresponding to the first order transfer function is not practically deployable. Hence, $\sigma \ge 0$ is essential and necessary to guarantee string stability. This completes the proof for Proposition 1.

Lemma 2*:* The proposed adaptive PD controller in Section 4.1 can ensure the string stability of a platoon if the time headway h and controller cut-off frequency $\omega_{K,i}$ satisfy the following conditions:

(i) For the CACC cases, the time headway h satisfies $h > 0$.

(ii) For the ACC case, a lower bound for $h\omega_{K,i}$ satisfies $h\omega_{K,i} \ge \sqrt{2}$.

Proof: The string stability can be analyzed by performing an arithmetic operation on the transfer function of string stability, by checking if the ratio of trajectory oscillations in the frequency domain of the ith vehicle and the leading vehicle 0 is lesser than or equal to 1.

(1) The CACC cases:

From Equations (40)-(42), the transfer functions of string stability for the CACC cases and the ACC case are:

$$\mathrm{SS}_{X,i} = \left(\alpha_{f,i}(\zeta_i)\Lambda_{f,i-1} + \alpha_{b,i}(\zeta_i)\Lambda_{b,i-1}\right)\frac{X_{i-1}}{X_0} + \left(\beta_{f,i}(\zeta_i)\Lambda_{f,i-2} + \beta_{b,i}(\zeta_i)\Lambda_{b,i-2}\right)\frac{X_{i-2}}{X_0} \tag{43}$$

Note that to directly analyze string stability transfer function is complex as it is a high-order transfer function. To address this problem, we consider only the worst case in Equation (43) where the values of both X_{i-2}/X_0 and X_{i-1}/X_0 are equal to one (implying head-to-tail marginally string stability, which means the traffic oscillations is neither amplified nor damped when it propagates in the traffic flow). This enables the determination of a more conservative, feasible region of the two parameters to ensure string stability. From Table 1, when $X_{i-2}/X_0 = X_{i-1}/X_0 = 1$, Equation (43) becomes:

$$\mathrm{SS}_{X,i} = \begin{cases} \left(\alpha G_i K_i + \beta G_i K_i + \alpha G_i F_{1,i} s^2 + \beta G_i F_{2,i} s^2\right)(1 + G_i K_i H_i)^{-1}, & CACC1 \\ \left(G_i K_i + G_i F_{1,i} s^2\right)(1 + G_i K_i H_i)^{-1}, & CACC2\ and\ CACC3 \end{cases} \tag{44}$$

Substituting for H_i from Equation (27), Equation (44) can be simplified to:

$$\mathrm{SS}_{X,i} = \begin{cases} [1 + (2-\alpha)hs]^{-1}, & CACC1 \\ (1 + hs)^{-1}, & CACC2\ and\ CACC3 \end{cases} \tag{45}$$

To guarantee the string stability condition in Equation (39), the magnitude of $\mathrm{SS}_{X,i}$ should not be greater than one. Correspondingly, for CACC1, according to proposition 1, the feasible region is: $h > 0$ since $1 \le 2 - \alpha < 2$ and time headway is positive; for CACC2 and CACC3, similarly, the feasible region is: $h > 0$.

(2) The ACC Case:

Under the ACC schematic, we have $\alpha_{f,i}(\zeta_i) = \beta_{f,i}(\zeta_i) = \beta_{b,i}(\zeta_i) = 0$, and $\alpha_{b,i}(\zeta_i) = 1$ from Table 1. The transfer function of string stability from Equation (43) will degrade to:

$$\mathrm{SS}_{X,i} = \frac{X_i}{X_0} = \Lambda_{b,i-1}\frac{X_{i-1}}{X_0} = \frac{G_i K_i}{1 + G_i K_i H_i} = \frac{\omega_{K,i} s + {\omega_{K,i}}^2}{\left(1 + h\omega_{K,i}\right)s^2 + \omega_{K,i}\left(1 + h\omega_{K,i}\right)s + {\omega_{K,i}}^2} \tag{46}$$

Consequently, by substituting $s = j\omega$ in Equation (49), the string stability condition becomes:

$$\left|\mathrm{SS}_{X,i}\right| = \left|\frac{j\omega_{K,i}\omega + {\omega_{K,i}}^2}{\left(\omega_{K,i} - \left(1 + h\omega_{K,i}\right)\omega^2\right) + j\omega_{K,i}\left(1 + h\omega_{K,i}\right)\omega}\right| \le 1 \tag{47}$$

Solving inequality (50) leads to:

$${\omega_{K,i}}^2\left(2 - h^2{\omega_{K,i}}^2\right)\left(1 + h\omega_{K,i}\right)^{-2} \le \omega^2 \tag{48}$$

Since $\min_{\omega \ge 0} \omega^2 = 0$, the inequality (48) can be solved by letting $\omega^2 = 0$. Then, the string stability region of the controller cut-off frequency and headway time is:

$$h\omega_{K,i} \ge \sqrt{2} \tag{49}$$

Remark 2: The decision-making process can be summarized as: (i) $h\omega_{K,i}$ has a specific upper bound for mitigating measurement noise effects; $h\omega_{K,i} \le W_{max} = 2$; (ii) string stability requires positive time headway: $h > 0$ for the CACC1, CACC2 and CACC3 cases, and $h\omega_{K,i} \ge \sqrt{2}$ for the ACC case; (iii) ensuring local stability (individual vehicle stability) requires positive time headway and controller cut-off frequency; and (iv) increasing either time headway or controller cut-off frequency will improve string stability performance though it deteriorates noise mitigation performance; hence, proper parameter selection is essential.

Remark 3: The additional parameter settings are: (i) the desired time headways h in all controllers should be identical to preclude traffic oscillations that are generated through controller switching; and (ii) to simplify the problem, this study considers a homogeneous platoon, implying that all the vehicles have the same adaptive PD controller.

5. Algorithm to solve OPT-II

The section describes the algorithm to solve the IFT optimization model OPT-II, formulated in Section 4.2. The objective function in Equation (17) is complex because: (i) the decision variable $\boldsymbol{\xi}$ is a binary vector, which makes it a discrete integer optimization problem; (ii) the transfer function $\mathrm{SS}^2_{X,i}(j\omega, \boldsymbol{\xi_d})$ depends on the controller used in the degeneration scenario $\boldsymbol{\xi_d}$, which implies different speed oscillation energies for the platoon under different degeneration scenarios; and (iii) the objective function requires the trajectory oscillations of the ambient traffic conditions in the frequency domain $X(j\omega)$ in the integrand to determine the expected speed oscillation energies. Together, these factors make it difficult to obtain a closed-form solution, entailing the need for a numerical solution.

To solve OPT-II numerically, the primary concern is computational complexity arising mainly due to platoon size. For a platoon with $N+1$ vehicles, there are 2^{N+1} candidate IFTs. For an IFT with b vehicles that activate "send"

functionalities, there are 2^b degeneration scenarios. For each degeneration scenario, the computation of the control performance requires the sum of the speed oscillation energies in the frequency domain for each vehicle. Hence, the computational complexity can increase significantly as the platoon size increases, leading to the need for computation efficiency. Though the optimal IFT is not updated frequently, it should be determined as quickly as possible so that the platoon can adjust the IFT to the optimal one before ambient traffic conditions change significantly. To enhance computation efficiency, we first analyze the activation statuses of the "send" functionalities of the leading and the last vehicles in the platoon as a pre-processing step, which can preclude the consideration of a subset of candidate IFTs. Then, a two-step algorithm is proposed to efficiently search for the optimal IFT among the remaining candidate IFTs.

5.1 Activation status of "send" functionalities of the leading and last vehicles in the platoon

The "send" functionality of the communication device for the last (following) vehicle should always be deactivated as it only needs to receive information to maintain control performance. Deactivating the "send" functionality of this vehicle can decrease the probability of communication collision and improve the reliability of the V2V communication.

Next, we investigate the activation status of the "send" functionality of the leading vehicle. We first compare the control performance of CACCs 1, 2, 3 and ACC using Proposition 2, and then prove that the leading vehicle always needs to activate its "send" functionality, using Lemma 3 and Theorem 1.

From Equation (43), we note that the transfer function of string stability $\mathrm{SS}_{X,i}$ of vehicle i is a function of its communication status ζ_i. In this subsection, we denote it more specifically as $\mathrm{SS}_{X,i,\zeta_i}$ to reflect the transfer function of string stability under different controllers (CACCs 1, 2, 3, and ACC, for $\zeta_i \in \{1,2,3,4\}$, respectively).

Proposition 2: Based on the four possible communication statuses in Fig. 4, the magnitudes of transfer functions of string stability satisfy: $\left|\mathrm{SS}_{X,i,1}\right| < \left|\mathrm{SS}_{X,i,2}\right| < \left|\mathrm{SS}_{X,i,4}\right|$, $\left|\mathrm{SS}_{X,i,1}\right| < \left|\mathrm{SS}_{X,i,3}\right| < \left|\mathrm{SS}_{X,i,4}\right|$, indicating that CACCs can damp traffic oscillations better than ACC.

Proof: To compare the magnitudes of string stability transfer functions under the CACCs and the ACC, the cut-off frequency ω_{c,ζ_i}, $\zeta_i \in \{1,2,3,4\}$ is analyzed. The cut-off frequency ω_{c,ζ_i} is a corner frequency beyond which the logarithmic value of X_i/X_0 should be smaller than -3.01dB (Palm, 2005) , which indicates that the oscillation energy will be effectively damped. A smaller cut-off frequency ω_{c,ζ_i} implies better string stability performance. The cut-off frequency ω_{c,ζ_i} of transfer function $\mathrm{SS}_{X,i,\zeta_i}(j\omega)$ can be calculated by solving Equation (50) as follows:

$$20\log\left|\mathrm{SS}_{X,i,\zeta_i}(j\omega_{c,\zeta_i})\right| = -3.01\text{dB} \tag{50}$$

From Equations (44) and (45), the string stability transfer function for CACC1 is:

$$\mathrm{SS}_{X,i,1}(j\omega_{c,1}) = (\alpha+\beta)(1+G_iK_iH_i)[H_i(1+G_iK_iH_i)]^{-1} = {H_i}^{-1} = \left[1+j(2-\alpha)h\omega_{c,1}\right]^{-1} \tag{51}$$

By substituting Equation (51) into Equation (50) and solving it, we obtain the corresponding cut-off frequency in Equation (52), where $C = 10^{\frac{-3.01}{10}}$:

$$\omega_{c,1} = \sqrt{(1-C)[(2-\alpha)^2Ch^2]^{-1}} \tag{52}$$

From Equations (44) and (45), the transfer functions for CACC2 and CACC3 are:

$$\mathrm{SS}_{X,i,2}(j\omega_{c,2}) = \mathrm{SS}_{X,i,3}(j\omega_{c,3}) = \left(G_iK_i + G_iF_{1,i}s^2\right)(1+G_iK_iH_i)^{-1} = {H_i}^{-1} = \left(1+j\omega_{c,2}h\right)^{-1} \tag{53}$$

By substituting Equation (53) into Equation (50) and solving it, we obtain the corresponding cut-off frequency:

$$\omega_{c,2} = \omega_{c,3} = \sqrt{(1-C)[Ch^2]^{-1}} \tag{54}$$

From Table 1, we have $0 < \alpha < 1$. By comparing Equations (52) and (54), we obtain that $\omega_{c,1} < \omega_{c,2} = \omega_{c,3}$. Then, we only need to compare $\omega_{c,2}$ or $\omega_{c,3}$ with $\omega_{c,4}$. From Equation (46), the transfer function of ACC is:

$$\mathrm{SS}_{X,i,4}(j\omega_{c,4}) = \frac{G_iK_i}{1+G_iK_iH_i} = \frac{j\omega_{K,i}\omega_{c,4} + {\omega_{K,i}}^2}{j\left({\omega_{K,i}}^2 + h\omega_{K,i}\right)\omega_{c,4} + {\omega_{K,i}}^2 - \left(1+h\omega_{K,i}\right){\omega_{c,4}}^2} \tag{55}$$

By substituting Equation (55) into Equation (50) and solving it, we obtain the corresponding cut-off frequency:

$$\omega_{c,4} = \sqrt{\left[B + \sqrt{B^2 - 4C\left(1+h\omega_{K,i}\right)^2\left({\omega_{K,i}}^4(C-1)\right)}\right]\left[2C\left(1+h\omega_{K,i}\right)^2\right]^{-1}} \tag{56}$$

where $B = C\left(\omega_{K,i} + h{\omega_{K,i}}^2\right)^2 - 2C{\omega_{K,i}}^2\left(1+h\omega_{K,i}\right) - {\omega_{K,i}}^2$.

From Remark 2, the values of $h\omega_{K,i}$ for controller design lie within the range $[\sqrt{2}, 2]$. From Equations (54) and (56), the necessary condition for the inequality $\omega_{c,4} > \omega_{c,2}$ is the lower bound of $h\omega_{K,i}$: $h\omega_{K,i} > (1+\sqrt{3})/2$. As the range of the controller parameters is included within the lower bound that is necessary for $\omega_{c,4} > \omega_{c,2}$, we can

conclude that ACC and CACC2 satisfy the condition $\omega_{c,4} > \omega_{c,2}$.

Based on the analysis above, we note that the cut-off frequencies of string stability transfer functions under the CACC cases and the ACC case satisfy $\omega_{c,1} < \omega_{c,2} = \omega_{c,3} < \omega_{c,4}$. Hence, the magnitudes of the string stability transfer functions satisfy: $|\mathrm{SS}_{X,i,1}| < |\mathrm{SS}_{X,i,2}| < |\mathrm{SS}_{X,i,4}|$, $|\mathrm{SS}_{X,i,1}| < |\mathrm{SS}_{X,i,3}| < |\mathrm{SS}_{X,i,4}|$.

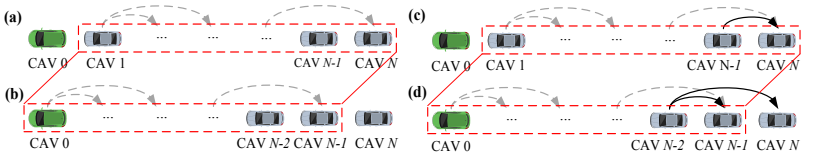


Fig. 6. Illustration of the deactivated or activated status of the "send" functionality of CAV 0.

Next, the "send" status of the leading vehicle is analyzed. Two IFTs are illustrated in Fig. 6(a) and 6(c). In both scenarios, the "send" functionality of the leading vehicle and the last vehicle are deactivated. In Fig. 6(a), vehicle $N-1$ has its communication device deactivated, while in Fig. 6(c) vehicle $N-1$ broadcasts messages. In Fig. 6(b) (Fig. 6(d)), the "send" activation status of vehicle i (for $i = 0,1,\cdots,N-1$) is the same as for vehicle $i+1$ in Fig. 6(a) (Fig. 6(c)). The "send" functionality is deactivated for the last vehicles in Fig. 6(b) and Fig. 6(d). All feasible IFT candidates can be categorized into these four general cases.

Lemma 3: In Fig. 6, the expected platoon speed oscillation energy of the IFT in Fig. 6(b) is always lower than that in Fig. 6(a). Similarly, the speed oscillation energy of the IFT in Fig. 6(d) is lower than that in Fig. 6(c).

Proof: Denote the IFTs in Figs. 6(a) and 6(b) as $\boldsymbol{\xi}^{(a)}$ and $\boldsymbol{\xi}^{(b)}$, and degeneration scenarios of $\boldsymbol{\xi}^{(a)}$, $\boldsymbol{\xi}^{(b)}$ as $\boldsymbol{\xi}_{\boldsymbol{d}}^{(a)}$, $\boldsymbol{\xi}_{\boldsymbol{d}}^{(b)}$. For each $\boldsymbol{\xi}_{\boldsymbol{d}}^{(a)}$, there is a $\boldsymbol{\xi}_{\boldsymbol{d}}^{(b)}$ whose IFT from vehicle 0 to vehicle $N-1$ is the same as the IFT of $\boldsymbol{\xi}_{\boldsymbol{d}}^{(a)}$ from vehicle 1 to vehicle N. Hence, the probability of $\boldsymbol{\xi}_{\boldsymbol{d}}^{(a)}$, $P_d(\boldsymbol{\xi}_{\boldsymbol{d}}^{(a)})$, is equal to the probability of $\boldsymbol{\xi}_{\boldsymbol{d}}^{(b)}$, $P_d(\boldsymbol{\xi}_{\boldsymbol{d}}^{(b)})$ (i.e., $P_d(\boldsymbol{\xi}_{\boldsymbol{d}}^{(a)}) = P_d(\boldsymbol{\xi}_{\boldsymbol{d}}^{(b)})$). Then, we only need to compare the control performance of $\boldsymbol{\xi}_{\boldsymbol{d}}^{(a)}$ and $\boldsymbol{\xi}_{\boldsymbol{d}}^{(b)}$. Note that vehicle 1 in Fig. 6(a) and vehicle N in Fig. 6(b) operate under ACC. To simplify notation, we use $\mathrm{SS}_{X,i,\zeta_i}(\boldsymbol{\xi}_{\boldsymbol{d}}^{(a)})$ and X_0 to denote $\mathrm{SS}_{X,i,\zeta_i}(j\omega,\boldsymbol{\xi}_{\boldsymbol{d}}^{(a)})$ and $X_0(j\omega)$, respectively. We denote $SS'_{X,i,\zeta_i}(\boldsymbol{\xi}_{\boldsymbol{d}}^{(a)}) = X_i/X_1$; then, according to Equations (11) and (38), we have:

$$E_d(\boldsymbol{\xi}_{\boldsymbol{d}}^{(a)}) = 4\pi^2 \int_0^{+\infty} \left(\sum_{i=0}^{N} \mathrm{SS}_{X,i,\zeta_i}^2(\boldsymbol{\xi}_{\boldsymbol{d}}^{(a)}) \right) \omega^2 X_0^2 d\omega = 4\pi^2 \int_0^{+\infty} \left(1 + \mathrm{SS}_{X,1,4}^2 \sum_{i=1}^{N} \mathrm{SS}'^2_{X,i,\zeta_i}(\boldsymbol{\xi}_{\boldsymbol{d}}^{(a)}) \right) \omega^2 X_0^2 d\omega \quad (57)$$

$$E_d(\boldsymbol{\xi}_{\boldsymbol{d}}^{(b)}) = 4\pi^2 \sum_{i=0}^{N} \int_0^{+\infty} \omega^2 SS_{X,i,\zeta_i}^2(\boldsymbol{\xi}_{\boldsymbol{d}}^{(b)}) X_0^2 d\omega = 4\pi^2 \int_0^{+\infty} \left(\sum_{i=0}^{N-1} SS_{X,i,\zeta_i}^2(\boldsymbol{\xi}_{\boldsymbol{d}}^{(b)}) + SS_{X,n,4}^2(\boldsymbol{\xi}_{\boldsymbol{d}}^{(b)}) \right) \omega^2 X_0^2 d\omega \quad (58)$$

$$\frac{E_d(\boldsymbol{\xi}_{\boldsymbol{d}}^{(b)}) - E_d(\boldsymbol{\xi}_{\boldsymbol{d}}^{(a)})}{4\pi^2} = \int_0^{+\infty} \left(\sum_{i=0}^{N-1} SS_{X,i,\zeta_i}^2(\boldsymbol{\xi}_{\boldsymbol{d}}^{(b)}) + SS_{X,n,4}^2(\boldsymbol{\xi}_{\boldsymbol{d}}^{(b)}) - 1 - \mathrm{SS}_{X,1,4}^2 \sum_{i=1}^{N} \mathrm{SS}'^2_{X,i,\zeta_i}(\boldsymbol{\xi}_{\boldsymbol{d}}^{(a)}) \right) \omega^2 X_0^2 d\omega \quad (59)$$

Since all vehicles have the same adaptive PD controller (Remark 2), we have $SS_{X,1,4}^2 = \Lambda_{b,0}^2 = \Lambda_{b,i-1}^2$ for any $i = 2,\ldots,N$, where $\Lambda_{b,i-1}^2$ is as shown in Equation (46). The receiver state ζ_i of vehicle i (for $i = 1,2,\cdots,N$) in Fig. 6(a) is equal to the receiver state ζ_{i-1} of vehicle $i-1$ in Fig. 6(b), which indicates that $\mathrm{SS}'^2_{X,i,\zeta_i}(\boldsymbol{\xi}_{\boldsymbol{d}}^{(a)}) = \mathrm{SS}_{X,i-1,\zeta_{i-1}}^2(\boldsymbol{\xi}_{\boldsymbol{d}}^{(b)})$. From Proposition 2 and Equations (38) and (46), for $i > 3$, we have $\mathrm{SS}_{X,i-1,\zeta_i}^2(\boldsymbol{\xi}_{\boldsymbol{d}}^{(b)}) \le \mathrm{SS}_{X,i-2,\zeta_{i-2}}^2(\boldsymbol{\xi}_{\boldsymbol{d}}^{(b)}) \Lambda_{b,i-2}^2 \le \mathrm{SS}_{X,i-3,\zeta_{i-3}}^2(\boldsymbol{\xi}_{\boldsymbol{d}}^{(b)}) \Lambda_{b,i-2}^2 \Lambda_{b,i-3}^2 \le \cdots < \mathrm{SS}_{X,1,\zeta_1}^2(\boldsymbol{\xi}_{\boldsymbol{d}}^{(b)}) \prod_{i'=2}^{i} \Lambda_{b,i'-1}^2 < \prod_{i'=1}^{i} \Lambda_{b,i'-1}^2 = \Lambda_{b,0}^{2i}$. For $i = 1,2$, we have $\mathrm{SS}_{X,i-1,\zeta_i}^2(\boldsymbol{\xi}_{\boldsymbol{d}}^{(b)}) < \prod_{i'=1}^{i} \Lambda_{b,i'-1}^2 = \Lambda_{b,0}^{2i}$. These inequalities indicate that the string stability performance of vehicle i under degeneration scenario $\boldsymbol{\xi}_{\boldsymbol{d}}^{(b)}$ is always better than the performance when all its predecessors deactivate their "send" functionalities. Accordingly, we have:

$$\sum_{i=0}^{N-1} SS_{X,i,\zeta_i}^2(\boldsymbol{\xi}_{\boldsymbol{d}}^{(b)}) + SS_{X,n,4}^2(\boldsymbol{\xi}_{\boldsymbol{d}}^{(b)}) - 1 - \mathrm{SS}_{X,1,4}^2 \sum_{i=1}^{N} \mathrm{SS}'^2_{X,i,\zeta_i}(\boldsymbol{\xi}_{\boldsymbol{d}}^{(a)}) \quad (60)$$

$$< \left(1-\Lambda_{b,0}^2\right)\left(\sum_{i=1}^{N-1}\Lambda_{b,0}^{2i}\right)+\Lambda_{b,0}^{2N}-1 \;=\left(1-\Lambda_{b,0}^2\right)\frac{1-\Lambda_{b,0}^{2N}}{1-\Lambda_{b,0}^2}+\Lambda_{b,0}^{2N}-1=0$$

By substituting inequality (60) into Equation (59), we have $E_d\left(\boldsymbol{\xi}_{\boldsymbol{d}}^{(b)}\right) < E_d\left(\boldsymbol{\xi}_{\boldsymbol{d}}^{(a)}\right)$, which indicates that each degeneration scenario $\boldsymbol{\xi}_{\boldsymbol{d}}^{(b)}$ of IFT $\boldsymbol{\xi}^{(b)}$ has lower oscillation energy than the corresponding degeneration scenario $\boldsymbol{\xi}_{\boldsymbol{d}}^{(a)}$ of IFT $\boldsymbol{\xi}^{(a)}$. Since the corresponding probabilities of each degeneration scenario pair are identical, $\boldsymbol{\xi}^{(b)}$ always outperforms $\boldsymbol{\xi}^{(a)}$. Similarly, we can prove that the IFT in Fig. 6(d) outperforms that in Fig. 6(c).

Theorem 1: In the optimal IFT, the "send" functionality of the leading vehicle is always activated.

Proof: If the "send" functionality of leading vehicle is not activated, then from Lemma 3 we can always change the IFT from (a) to (b) or from (c) to (d) to find a better one until the leading vehicle activates the "send" functionality.

5.2 Two- step algorithm for solving OPT-II

Different IFTs may have several identical degeneration scenarios. For example, IFT $[0, 0, 1, 0, 0]$ in Fig. 2(a) is a degeneration scenario of IFT $[1, 0, 1, 0, 0]$ illustrated in Fig. 2(a). However, it is also a degeneration scenario of IFT [1, 1, 1, 0, 0] when both vehicles 0 and 1 fail to send messages at the same time. According to Theorem 1 in Section 5.1, the leading vehicle always needs to activate its "send" functionality and the last vehicle needs to deactivate it, a fully-activated IFT except for the last vehicle ($\boldsymbol{\xi} = [1, \ldots, 1,0]$) includes all possible degeneration scenarios for other IFTs. Therefore, we only need to investigate the string stability performance of the degeneration scenarios for that IFT. For the other IFTs, we will just use the string stability performance of the relevant degeneration scenarios calculated for the fully-activated IFT.

Motivated by above observation, we propose a two-step algorithm. The first step calculates the string stability performance of degeneration scenarios for the fully-activated IFT to construct a control performance table according to Sections 3.1 and 4. Then, in the second step, for each $\boldsymbol{\xi} \in \boldsymbol{\Omega}$, we traverse all possible degeneration scenarios $\boldsymbol{\xi_d} \in \boldsymbol{\Omega_d}(\boldsymbol{\xi})$, and add the corresponding control performances from the table generated in the first step with a weight $P_d(\boldsymbol{\xi_d}(\boldsymbol{\xi}))$ formulated from the contention model in Section 3.2 to obtain the expected string stability of IFT $\boldsymbol{\xi}$. The pseudo code of the two-step algorithm is shown in the following table:

Step 1

input ambient traffic oscillations in the frequency domain $X(j\omega)$, average density $\bar{k}$, and platoon size $N+1$;
set $\boldsymbol{\xi} = [1, \ldots, 1,0] \in \mathbb{R}^{N+1}$, update $\boldsymbol{\Omega_d}(\boldsymbol{\xi})$
for any $\boldsymbol{\xi_d}(\boldsymbol{\xi}) \in \boldsymbol{\Omega_d}(\boldsymbol{\xi})$
  Determine its speed oscillation energies $E_{\boldsymbol{d}}\left(\boldsymbol{\xi_d}(\boldsymbol{\xi})\right)$ using Equation (11).
end
output $\hat{\boldsymbol{\xi}}_{\boldsymbol{d}} = \boldsymbol{\xi_d}(\boldsymbol{\xi})$ and corresponding speed oscillation energies $\hat{E}_{\boldsymbol{d}}(\hat{\boldsymbol{\xi}}_{\boldsymbol{d}}) = E_{\boldsymbol{d}}\left(\boldsymbol{\xi_d}(\boldsymbol{\xi})\right)$ for any $\boldsymbol{\xi_d}(\boldsymbol{\xi}) \in \boldsymbol{\Omega_d}(\boldsymbol{\xi})$

Step 2

input *E(x)*
set the set of candidate IFTs $\boldsymbol{\Omega} = \{[1, \eta_1, \ldots, \eta_{N-1}, 0] | \eta_i \in \{0,1\} \text{ for } i = 1, \ldots, N-1\}$
initialize the optimal expected speed oscillation energies $E^* \leftarrow +\infty$
for any $\boldsymbol{\xi} \in \boldsymbol{\Omega}$
  Update $\boldsymbol{\Omega_d}(\boldsymbol{\xi})$
  for any $\boldsymbol{\xi_d}(\boldsymbol{\xi}) \in \boldsymbol{\Omega_d}(\boldsymbol{\xi})$
    Determine the $P_d\left(\boldsymbol{\xi_d}(\boldsymbol{\xi})\right)$ using Equation (16).
    Find $\hat{\boldsymbol{\xi}}_{\boldsymbol{d}} = \boldsymbol{\xi_d}(\boldsymbol{\xi})$, then $E_{\boldsymbol{d}}\left(\boldsymbol{\xi_d}(\boldsymbol{\xi})\right) \leftarrow \hat{E}_{\boldsymbol{d}}(\hat{\boldsymbol{\xi}}_{\boldsymbol{d}})$.
  end
  Determine the expected speed oscillation energies $E(\boldsymbol{\xi})$ under the IFT $\boldsymbol{\xi}$ using Equation (2).
  if $E(\boldsymbol{\xi}) < E^*$
    Update $E^* \leftarrow E(\boldsymbol{\xi})$ and the optimal IFT $\boldsymbol{\xi}^* \leftarrow \boldsymbol{\xi}$
  end
end
output $\boldsymbol{\xi}^*$

6. Numerical experiments

6.1 Experiment design and parameter setting

Numerical experiments are conducted to analyze the CACC-OIFT strategy. First, the performance of V2V communication and the computational efficiency of the algorithm are investigated. Next, the performance of CACC-OIFT is analyzed. The optimization procedure and the performance comparison simulations are conducted on a C++ platform that integrates network simulator NS-3 to emulate the V2V communication process.

The experiment setup consists of a $N+1$ CAV platoon with one leading vehicle ($i=0$) and N following vehicles ($i=1,\dots,N$, and $N=11,\dots,15$). The movement of the leading vehicle is predetermined according to NGSIM field data (US DOT, 2007), which contains a 240-second vehicle trajectory on eastbound I-80 in the San Francisco Bay area at Emeryville, California. The frequency domain trajectory oscillations $X(j\omega)$ and average density of ambient traffic flow $\bar{k}$ are provided to the optimization model. The first following vehicle ($i=1$) receives information only from one preceding vehicle ($i=0$); so, the controller will switch between CACC2 and ACC if the IFT degenerates. For the other vehicles ($i=2,\dots,N$), the controller can switch among the four controllers (i.e., CACC 1, 2, 3, and ACC). The desired time headways in all controllers are set to $h=1s$. The cut-off frequency $\omega_{K,i}$ is set as 0.8, 0.8, 0.9 and 1.45 for CACC 1, 2, 3, and ACC, respectively. The parameters α and β are set as 0.7 and 0.3. The control time interval is set as 0.1s. The network parameters are listed in Table 2. In addition, since the measurement noise generated from onboard sensors and vehicle movements cannot be neglected in CACC/ACC implementation, we include the measurement noise in the measured position and speed of predecessors. The position measurement noise and speed measurement noise are both Gaussian white noise. As illustrated in (Moon et.al., 2005), the standard deviation (SD) of the speed measurement noise is set as: $\sigma_v=0.1$m/s; and according to (Kim, 2012), the SD of position measurement noise can be calculated as: $\sigma_x=\frac{T\sigma_v}{\sqrt{2}}=0.1\times\frac{0.1}{\sqrt{2}}=0.007$m.

Table 2. Network parameters.

Parameter	Value	Parameter	Value	Parameter	Value
Communication range	0.2 km	Information generation rate	10 Hz	Data rate	3 Mbps
Packet size	500 B	Contention window size	8	Slot time	16 μs

The performance of CACC-OIFT is investigated through three numerical experiments. The first experiment compares the proposed CACC-OIFT with two other control strategies. The three strategies are: (i) CACC-OIFT, which includes the IFT optimization from Section 3 and the adaptive PD controller from Section 4, (ii) CACC-DIFT, which includes the adaptive PD controller from Section 4 with a fully-activated IFT, and (iii) CACC with a fixed IFT (CACC-FIFT), which includes the CACC and ACC schemes developed in Naus et.al. (2010). We also analyze another CACC controller with a fixed IFT (Schakel et.al., 2010). However, as its string stability is not guaranteed, it performs worse than the CACC in Naus et.al. (2010). Hence, hereafter, we focus on the CACC-FIFT developed by Naus et.al. (2010) for comparison purposes. In the second experiment, we provide an optimal IFT updating scenario to study the transition process of switching optimal IFT when the platoon size changes. Additionally, as the controller cut-off frequency $\omega_{K,i}$ has a significant impact on the performance of platoon control, the third experiment performs sensitivity analysis to illustrate the impact on system performance. The scenarios for control parameters $\omega_{K,i}$ are labeled as scenarios 1 to 3 in Table 3, in which we sequentially increase the value of $\omega_{K,i}$ for the four controller sets simultaneously.

Table 3. Values of $\omega_{K,i}$ in the sensitivity analysis of controller cut-off frequency.

Scenario	1	2	3
CACC1	0.8	0.96	1.152
CACC2	0.8	0.96	1.152
CACC3	0.9	1.08	1.296
ACC	1.45	1.74	2.088

6.2 Performance of V2V communications, optimization result and computation efficiency

To illustrate the performance of V2V communications under different ambient traffic conditions, we conduct simulations with different parameters. Fig. 7 shows the communication success rates under different ambient traffic conditions and different IFTs from the simulations in NS-3. In Fig. 7(a), the x-axis denotes the average traffic density $\bar{k}$. This study sets the range of average density from 25 vehicles/km to 40 vehicles/km. The success rate decreases with the increase in the percentage of vehicles with activated “send’ functionalities of V2V communication devices

in communication range. A higher percentage of activated "send" functionalities leads to more intense contentions for the chance to broadcast. For the same proportion of activated "send" functionalities in communication range, a higher traffic density will result in a higher failure probability. This is because there are more vehicles in the communication range of each vehicle if the average density of ambient traffic flow is higher. Hence, there are more vehicles with activated "send" functionalities when the proportions are identical. Fig. 7(b) compares the simulation results under $\bar{k} = 25$ with the contention model (discussed in Section 3.2) that is calibrated for the IFT optimization model. The mean error is -0.25% and the standard deviation of the error is 0.0526. Similar results are observed under different average densities, implying that the model can accurately describe the success rate under different percentages of activated "send" functionalities. We do not show all results due to the page limit.

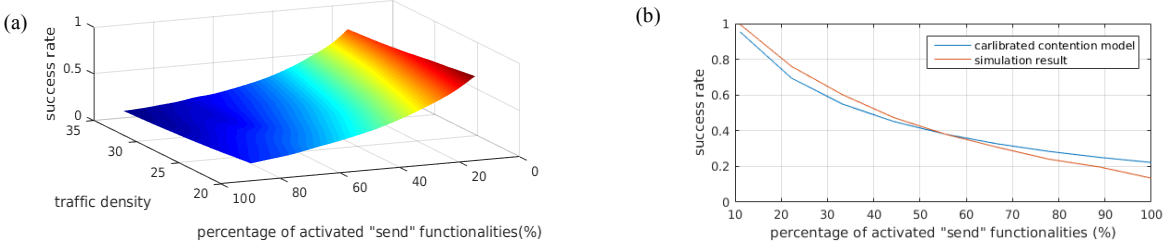


Fig. 7. Experiment results for V2V communications: (a) Communication success rates under different ambient traffic conditions; (b) Comparison of calibrated contention model and simulation results for k=20.

Table 4 illustrates the optimal IFTs under different ambient traffic conditions and platoon sizes. It can be observed that the optimal IFTs are in accordance with Theorem 1 and Corollary 1. Consecutive vehicles with activated "send' functionalities can efficiently increase the occurrence of CACC cases, especially the CACC1 case which has much better control performance than the ACC case. However, the success rate of communication decreases since it increases the probability of information collision. Thereby, there are some consecutive vehicles with deactivated communication devices directly following those with activated ones, such as vehicles 3, 4, and 5 for the scenario with $\bar{k}$=25 and N=14. For a given platoon size, the number of consecutive vehicles with deactivated "send' functionalities is bigger if the density $\bar{k}$ is higher. For a given $\bar{k}$, there exist several activated/deactivated patterns of communication devices in the platoon. For example, the same pattern 111000 exists for the first 6 and last 6 vehicles in the scenario $\bar{k}$=25, and N=14, as the communication environment and control scheme are similar for a pattern.

Table 4. Network parameters.

N=14	Optimal IFT	$\bar{k}$=25	Optimal IFT
$\bar{k}$=25	111000111000110	N=11	111000111000
$\bar{k}$=30	111000001110000	N=12	1110001111000
$\bar{k}$=35	110000001110000	N=13	11100011110000
$\bar{k}$=40	110000000110000	N=14	111000111000110
		N=15	1110001110001100

To enable practical deployment of CACC-OIFT, computational efficiency should be verified. Though we improve the algorithmic efficiency in several parts as discussed in Section 5, it is a brute-force method; the computational time increases with platoon size. However, this is not an issue in our study as the platoon length cannot be too long because: (1) long platoons will block other vehicles from changing lanes (Van Arem et al., 2006), and (2) regulatory policies can limit platoon length in the real world to ensure safe travel experience for all vehicles. In our experiments, the optimization procedure for a platoon with 15 vehicles only takes 48.23 seconds on a PC with Intel E3-1505M 2.80GHz 8Gb. Further, in practice, parallel computing can be leveraged. Then, the computational time for this platoon reduces to 3.22 seconds. Since the optimal IFT is updated every time period (e.g., 5 minutes) or when ambient traffic oscillation conditions change significantly, this small computation time ensures the practical applicability of CACC-OIFT.

6.3 Control performance evaluation

The study evaluates the performance of the proposed CACC-OIFT through three numerical experiments, in which we compare the control performance to the other two controllers (CACC-FIFT and CACC-DIFT), investigate the

transition process when the optimal IFT is updated, and illustrate the impact of cut-off frequency $\omega_{K,i}$ on platoon control performance.

6.3.1. Comparison between controllers

Here, we first compare the performance of CACC-OIFT with those of CACC-DIFT and CACC-FIFT. The experiments analyze the three controllers in the context of unreliable V2V communications by simulating in NS-3. A 15-CAV platoon is analyzed in a traffic flow with average density 28.57 vehicle/km for 240s. The ambient traffic conditions do not change significantly. Under CACC-OIFT, the vehicle platoon will follow the IFT from the optimization model (111000111000110). Fig. 8 shows the spacing speed tracking error between adjacent vehicles in the platoon under these controllers, and Fig. 9 shows the standard deviations of the spacing and speed tracking errors, and vehicle speed.

Fig. 8 illustrates that the spacing error of vehicles is mitigated based on their positions in the platoon. The figure shows that CACC-OIFT outperforms the other two controllers. For example, the maximum spacing error of the second following vehicle (*i=2*) under CACC-OIFT is 1.05m, compared with 1.42m for CACC-DIFT and 1.51m for CACC-FIFT. For the last following vehicle (*i=14*), the maximum spacing errors are 0.37m, 0.68m and 0.79m for CACC-OIFT, CACC-DIFT and CACC-FIFT, respectively. The standard deviations of spacing and speed tracking errors are compared for the three controllers in Fig. 9(a) and Fig. 9(b), respectively. Fig. 9(a) shows that the standard deviation of spacing error decreases sequentially across vehicles in the platoon for all controllers. However, CACC-OIFT performs better than the other two controllers as its spacing error reduces the quickest. Further, the profile of the spacing error standard deviation cycles from steep to flat. For example, the spacing error is reduced significantly for first 4 vehicles and then is almost constant for vehicles *i=5* and *i=6*. This is because the IFT optimization deactivates the "send' functionalities of V2V communication devices for several vehicles. Thus, some vehicles will operate under the ACC case, which does not leverage V2V connectivity. However, these deactivations lead to more reliable V2V communication connections for remaining links in CACC-OIFT and the consequent significant tracking error reduction for the vehicles. A similar trend is observed in Fig. 9(b) which shows the standard deviation of the speed tracking error.

To further investigate the performance benefits under CACC-OIFT, the performance of the three CACC control strategies is compared when traffic oscillates (e.g., stop-and-go or slow-and-fast traffic). The standard deviations of the vehicle speed are shown in Fig. 9(c). It illustrates that the fluctuation in standard deviation of speed decreases under all three schemes as the tail of the platoon is approached, which implies that traffic oscillations are damped. Further, CACC-OIFT reduces the speed fluctuations significantly as it proactively leverages the dynamic nature of the IFT. In summary, we conclude that the performance of a CAV platoon controlled by the proposed CACC-OIFT is better and more robust than that of the other two controllers in a realistic V2V communications environment. Also, based on the discussion in Section 1, CACC-DIFT performs better than CACC-FIFT as it considers IFT dynamics, albeit passively, unlike CACC-FIFT which assumes a fixed IFT.

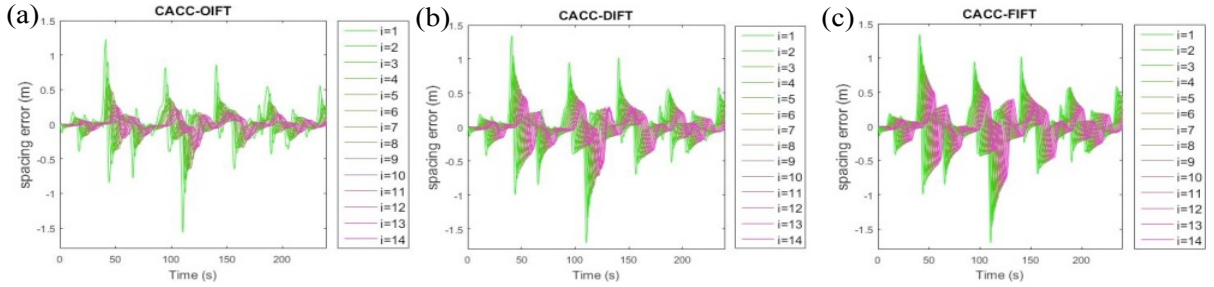


Fig. 8. Spacing error under different controllers: (a) CACC-OIFT; (b) CACC-DIFT; (c) CACC-FIFT.

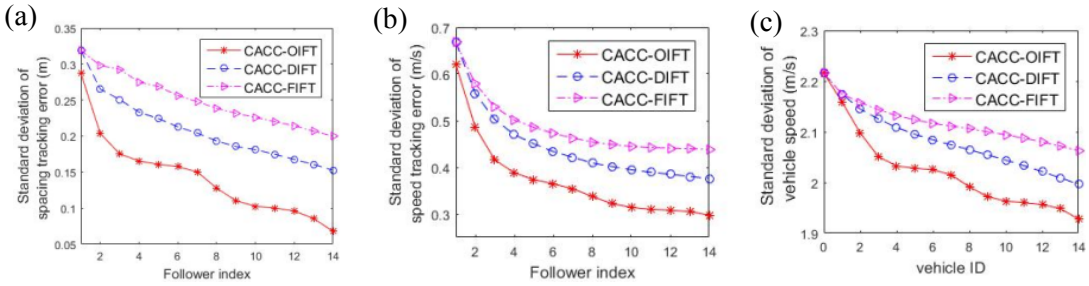


Fig. 9. Standard deviation of: (a) spacing error; (b) speed tracking error; (c) vehicle speed.

6.3.2. Transition process of optimal IFT

Next, we analyze the transition process of switching optimal IFT. In this experiment, a 14-CAV platoon is initially set up and controlled using CACC-OIFT with the optimal IFT from the optimization model (11100011110000). In the middle of the experiment (in time), a CAV approaches the tail of the platoon. At the 120th second, its headway becomes 1.5s, and this CAV joins the platoon (it terminates individual control mode and starts to apply the CACC-OIFT of the platoon). Then, as the platoon size changes from 14 to 15, the optimal IFT is updated to 111000111000110. Fig. 10 shows the spacing error, the speed, and the acceleration for all CAVs.

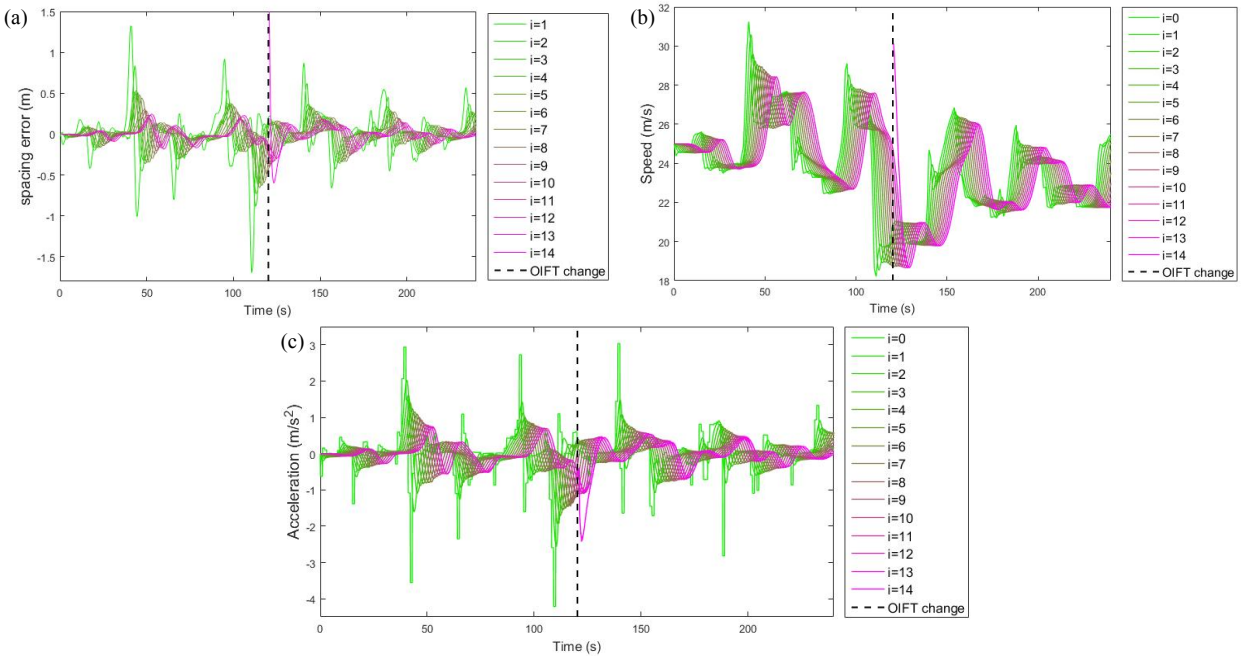


Fig. 10. Performance of the platoon in IFT transition period: (a) spacing error; (b) speed; (c) acceleration.

The results illustrate that the proposed CACC-OIFT is not affected by the transition of the optimal IFT. Specifically, in Fig. 10(a), the preceding 13 CAVs can still follow their leading vehicles with stable spacing errors after the last following CAV (*i=14*) joins in the platoon. The spacing error is also mitigated from the head to the tail of the platoon before and after the transition. The transition does not trigger any additional speed oscillations or acceleration oscillations for the platoon, as shown in Figs. 10(b) and 10(c). The last following CAV (*i=14*) decelerates appropriately to join the platoon with the desired headway (i.e., 1s). In this process, the maximum spacing error is 0.51m, and the maximum acceleration rate is -2.58m/s^2. The reason for the maximum acceleration rate of the last following CAV (*i=14*) being greater than that of its immediate predecessor CAV (*i=13*) is because the relative speed is almost -5m/s when the last CAV joins in the platoon. Thereby, it has to decelerate abruptly. After the join-in process

(i.e., after 145s), the maximum acceleration rate of the last following CAV (*i=14*) becomes smaller than that of its immediate predecessor, and the string stability performance is maintained. This indicates that the transition of optimal IFT induced by a change in platoon size through the addition of another CAV does not impact the performance of CACC-OIFT.

6.3.3. Sensitivity analysis of cut-off frequency

The sensitivity analysis of the cut-off frequency $\omega_{K,i}$ is conducted by comparing the spacing errors and vehicle acceleration in Scenarios 1, 2 and 3, as shown in Figs. 11 and 12, respectively. Fig. 11 illustrates that spacing error will decrease as the value of $\omega_{K,i}$ increases. For example, the maximum spacing error of the last following vehicle in Scenarios 1, 2 and 3 are 0.37m, 0.32m, and 0.24m, respectively. However, Fig. 12 shows that with the increase of $\omega_{K,i}$, the effect of high-frequency measurement noise will increase in the acceleration profile as it reaches the tail of the platoon (such as the parts marked by red circles in Fig. 12 (b)), which is undesirable from the perspective of passengers' comfort and vehicle operation.

Insights into the above observations can be generated by characteristics related to string stability performance and measurement noise mitigation. From the standpoint of string stability performance, the string stability transfer functions in Equations (45) and (47) imply that increasing the controller cut-off frequency $\omega_{K,i}$ will decrease the value of string stability transfer function, improving the damping effect of traffic oscillations. However, for the noise mitigation effect, the complementary sensitivity functions in Equations (36) and (37) indicate that as the value of $\omega_{K,i}$ becomes larger, the noise mitigation factor will increase, degrading the noise mitigation performance. Hence, increasing controller cut-off frequency will jeopardize the control performance related to noise mitigation. Hence, to enable an acceptable level of platoon control performance, a careful selection of the controller cut-off frequency $\omega_{K,i}$ is essential for real-world application.

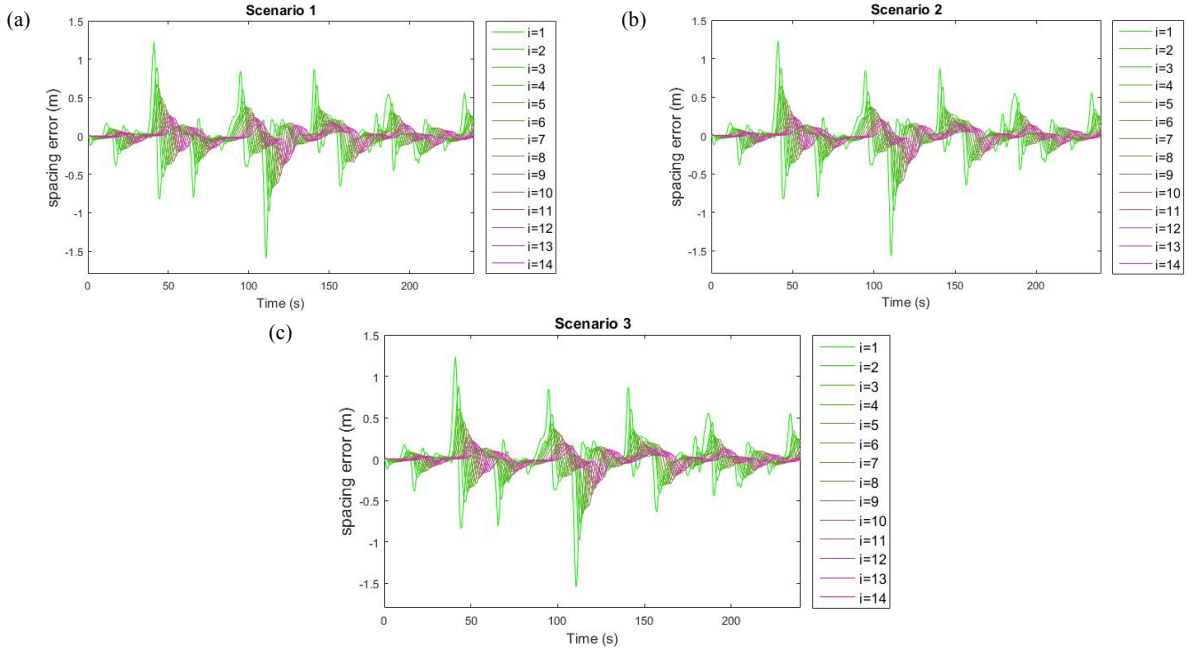


Fig. 11. Spacing error profile of: (a) Scenario 1; (b) Scenario 2; (c) Scenario 3.

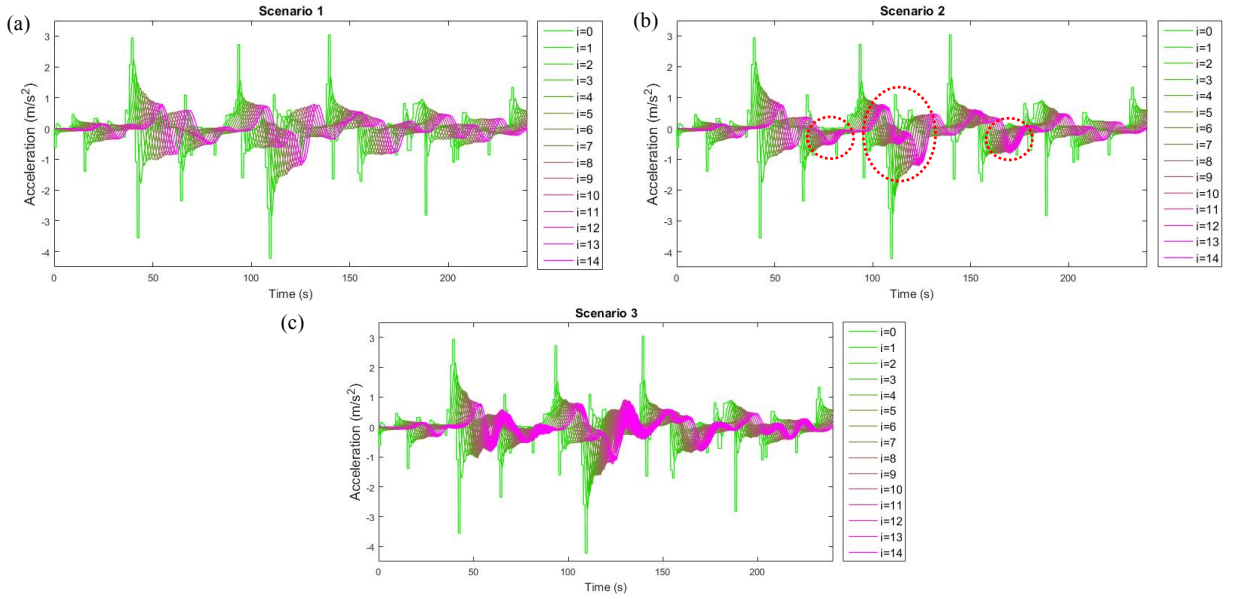


Fig. 12. Acceleration profile of: (a) Scenario 1; (b) Scenario 2; (c) Scenario 3.

7. Concluding comments

This study proposes a novel CACC strategy, CACC-OIFT, to explicitly factor IFT dynamics and to leverage it to enhance the platoon performance in an unreliable V2V communication context for a pure CAV platoon. The proposed CACC-OIFT consists of an adaptive PD controller and an IFT optimization model. Given the adaptive PD controller with a two-predecessor-following scheme, and the ambient traffic conditions and the platoon size just before the start of a time period, the IFT optimization model determines the optimal IFT that dynamically activates and deactivates the "send" functionality of the V2V communication devices of all vehicles in platoon, which maximizes the expected string stability. Since communication failures can cause IFT to degenerate dynamically, all possible degeneration scenarios for that IFT are considered in this expectation. The degeneration scenario probabilities are determined based on the communication failure probabilities for that time period which depend on the ambient traffic conditions. In the operational deployment context, based on the various degeneration scenarios for the optimal IFT at different time instants within the time period, the adaptive PD controller continuously determines the car-following behaviors of the vehicles in the platoon. A two-step algorithm is proposed to solve the IFT optimization problem by leveraging some key proven properties, such as the leading vehicle in the platoon should always activate its "send" functionality. Extensive numerical simulations are conducted in NS-3 to illustrate the effectiveness of CACC-OIFT.

To the best of our knowledge, this is the first attempt to explicitly factor IFT dynamics and to leverage it to enhance the performance of CACC strategies. Further, it is the first study to perform a rigorous mathematical modeling of the problem to theoretically illustrate properties. The insights from numerical experiments suggest that CACC-OIFT can leverage IFT dynamics to proactively reduce V2V communication failures while ensuring realism in terms of factoring the ambient traffic conditions. Further, the proposed two-step algorithm and its ability to be parallelized ensure computational tractability for operational deployment for platoons of considerable size (15 vehicles in this study). Also, the study insights provide key pointers for future CACC designs, in that communication failures and IFT dynamics should be considered to enable realism and enhance control performance. In summary, CACC-OIFT can generate a more reliable IFT for a CAV platoon, damp traffic oscillation propagation, and stabilize the traffic flow more efficiently for the entire platoon. Thereby, CACC-OIFT is string stable and outperforms strategies proposed in the current literature, CACC-DIFT and CACC-FIFT, considerably.

The proposed modeling approach provides opportunities for improvements beyond the current study. In the V2V communications context, our future work includes: (i) considering the role of receiver failure; (ii) factoring retransmission mechanism in modeling process; and (iii) investigating the application of other communication protocols (e.g., 5G). In the context of controller design, this research motivates the following tasks: (i) factoring communication delay, actuator delay, nonlinear vehicle dynamics and external disturbances to design a more realistic

controller; (ii) modeling heterogeneous vehicle platoons, and deriving the heterogeneous string stability condition; (iii) analyzing stability of a hybrid dynamic system during the switching process; and (iv) analyzing the switching process to guarantee smooth transition between controller sets. Related to the optimization problem, possible future directions include: (i) including more performance matrices or constraints (e.g., comfort, fuel consumption) into the objective function; and (ii) developing a more efficient algorithm to reduce computational time.

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